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IMPROVED PROTEIN PURIFICATION

Abstract

The present invention relates to protein purification, primarily in the chromatographic field. More closely, the invention relates to affinity chromatography using a split intein system comprising a C-intein tag and N-intein ligand, wherein the N-intein ligand provides increased solubility suitable for large scale purification of any recombinant target protein.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to protein purification, primarily in the chromatographic field. More closely, the invention relates to affinity chromatography using a split intein system comprising a C-intein tag and N-intein ligand, wherein the N-intein ligand has high solubility and may be immobilized to a solid phase in high degree suitable for large scale protein purification.

BACKGROUND OF THE INVENTION

[0002] Inteins are protein elements expressed as in-frame insertions that interrupt enzyme sequences and catalyze their own excision and ligation of two flanking polypeptides, generating an active protein. Genetically, inteins are encoded in two distinct ways: as intact inteins, interrupting two flanking extein sequences, or as split inteins, wherein each extein and part of the intein are encoded by two different genes. While they hold great promise as bioengineering and protein purification tools, split inteins with rapid kinetic properties found in nature are dependent on specific amino acids at the intein-extein junction, severely limiting the proteins that can be fused to inteins for affinity purification and recovery of native protein sequences. In particular, the prototypical split intein DNAE from *Nostoc punctiforme* exhibits kinetic properties suitable for protein purification applications. However, its activity is dependent on phenylalanine at the +2 position in the C-extein. This dependency severely narrows and impairs its general applicability.

[0003] Inteins have been engineered to accomplish several important functions in biotechnology, including applications as self-cleaving proteins for recombinant protein purification. Split inteins are particularly promising in this regard, as they can simultaneously provide affinity ligand and self-cleavage properties. In protein purification, a target protein that is the subject of purification may be substituted for either extein. To date, the DNAE family of split inteins has shown the most promise with C-terminal cleavage protein purification approaches.

[0004] WO2014/004336 describes proteins fused to split intein N-fragments and split intein C-fragments which could be attached to a support. The solid support could be a particle, bead, resin, or a slide.

[0005] WO2014/110393 describes proteins of interest fused to a split intein C-fragment which is contacted with a split intein N-fragment and a purification tag. The N-fragment may be attached to a solid phase via the purification tag and methods for affinity purification are discussed.

[0006] U.S. Pat. No. 10,066,027 describes a protein purification system and methods of using the system. Disclosed is a split intein comprising an N-terminal intein segment, which can be immobilized, and a C-terminal intein segment, which has the property of being self-cleaving, and which can be attached to a protein of interest. The N-terminal intein segment is provided with a sensitivity enhancing motif which renders it more sensitive to extrinsic conditions.

[0007] U.S. Pat. No. 10,308,679 describes fusion proteins comprising an N-intein polypeptide and N-intein solubilization partner, and affinity matrices comprising such fusion proteins.

[0008] WO 2018/091424 describes a method for production of an affinity chromatography resin comprising an amino-terminal, (N-terminal), split intein fragment as an affinity ligand, comprising the following steps: a) expression of an N-terminal split intein fragment protein as insoluble protein in inclusion bodies in bacterial cells, preferably *E. coli*, b) harvesting said inclusion bodies; c) solubilizing said inclusion bodies and releasing expressed protein; d) binding said protein on a solid support; e) refolding said protein; f) releasing said protein from the solid support; and g) immobilizing said protein as ligands on a chromatography resin to form an affinity chromatography resin. This procedure enables immobilization a ligand density of 2-10 mg/ml resin.

[0009] As described above, split inteins have been used for protein purification using a combined affinity tag and tag cleavage mechanism. However, the utility of such systems, is limited by several factors. First, there is the amino acid requirements at the splice junction of the intended product, i.e. the requirement of Phe in the +2 position of the C-extein, to effect cleavage and attain purification of tag-less proteins. Recombinant protein production without extraneous amino acid on the N-

terminus is highly desirable. Second, the protein releasing cleavage has to be sufficiently fast and provide an acceptable yield. Third, there is a solubility requirement of the split intein N- or C-fragment for attachment thereof to a solid support. Fourth, hitherto there are no available split intein systems suitable for large scale purification of tag-less proteins.

[0010] One way to improve the solubility is by attaching a solubility fusion-tag to the split-inteins, (U.S. Pat. No. 10,308,679). The development of methods for protein expression and purification is commonly facilitated by the use of fusion tags that offer the possibility to standardize protocols for purification, simplify the detection and increase the solubility of a target protein. Fusion tags can however interfere with protein function and structure. It is therefore advantageous to remove fusion tags prior to usage. A large fusion tag relative to a target protein also results in an increased metabolic burden for the host cells expressing these fusion proteins, since additional energy is spent on the fusion tag.

[0011] A different approach to increase the efficiency of producing highly insoluble split-inteins is by solubilizing proteins with denaturing chemical reagents followed by a refolding process, (U.S. application Ser. No. 16/348,534) to regain bioactive protein. Attempts have been made to understand the technical aspects of various methods used for protein refolding along with their advantages and limitations, but usually the efficiency and yield in such methods is very difficult to predict and has to be determined by empirical studies for each protein. A common problem in refolding methods is the formation of protein aggregates when the denaturing chemicals are being removed or diluted during refolding. These aggregates lower the yield in the process and adds complexity during the subsequent purification steps in a production process. Moreover, the denaturing chemicals are usually a burden to the environment and needs to be properly handled.

SUMMARY OF THE INVENTION

[0012] The present invention overcomes the disadvantages within prior art and provides a N-intein polypeptide, which is soluble without the need for a solubility fusion-tag and that can be produced in an industrial scale in an environmentally friendly production process for subsequent use in affinity purification processes.

[0013] In particular, the invention provides a method to increase the solubility of the prototypical split intein DnaE from *Nostoc punctiforme*, which exhibits kinetic properties suitable for protein purification applications. However, the method is not limited to DnaE split intein from *Nostoc punctiforme* but is also applicable for homologous split inteins from other species. Solubility refers to a protein that after a substitution of one or preferably two amino acids in the polypeptide chain has a higher ratio of soluble N-intein expressed in *E. coli* relative to the soluble N-intein ratio in the absence of these amino acid substitutions.

[0014] The present invention provides N-intein protein variants of native split inteins or consensus sequences derived from inteins/split inteins wherein the N-intein protein variant has one or more mutations for increased solubility.

[0015] In a first aspect, the invention relates to an N-intein variant derived from native *Nostoc punctiforme* (Npu) or sequences having at least 95% homology therewith comprising at least one amino acid substitution of a native split intein wherein the N-intein protein variant sequence includes a mutation in at least position 24 and/or position 25 as measured from the initial catalytic cysteine and wherein the substituted amino acid provides increased solubility in aqueous buffers compared to the native N-intein protein sequence or a consensus N-intein sequence. The invention also encompasses inteins which have a naturally occurring E in position 24, such as N-inteins from *Limnoraftis robusta*.

[0016] Preferably the substituted amino acid(s) that provide increased solubility is a non-positive amino acid. In a preferred embodiment the substituted amino acid that provide increased solubility is K24E. In another embodiment the substituted amino acid that provide increased solubility is R25N. In a most preferred embodiment the N-inten comprises both these mutations.

[0017] The invention relates to an N-intein protein variant of the wildtype N-intein domain of

Nostoc punctiforme (Npu) wherein the wildtype Npu N-intein domain comprises the following sequence:

CLSYETEILTVEYGLLPiGKIVEKRIECTVYSVDNNGNIYTQPVAQWHDRGEQEVFEY
CLEdGSLIRATKdHKFMTVDGQMLPiDEIFERELdLMRV (SEQ ID NO 1) (construct A52 in the Examples), wherein the protein variant comprises an amino acid substitution from K to E in position 24 and R to N in position 25 (construct B97 in the Examples) to increase solubility in aqueous buffers to minimize formation of inclusion bodies, wherein optionally one or more C (Cys) are mutated to non-Cystein residues, preferably S (Ser) or A (Ala). Further constructs encompassed by the invention are described in the example section below.

[0018] The N-intein protein variant as described above has solubility in aqueous buffer of at least 10-40% soluble N-intein with a single-point mutation of R at position 25, preferred N or non-positive amino acid; at least 46-52% soluble N-intein with a single-point mutation of K at position 24, preferred E or non-positive amino acid; and at least 76-88% soluble N-intein with mutations at positions 24 and 25, preferred K24E and R25N or non-positive amino acids.

[0019] The N-intein variant may be coupled to solid phase, such as a membrane, fiber, particle, bead or chip, such as chromatography resin of natural or synthetic origin.

[0020] The solid phase may optionally be provided with embedded magnetic particles. In an alternative embodiment the solid phase is a non-diffusion limited resin/fibrous material. According to the invention 0.2-2 μ mole/ml N-intein is coupled per ml solid phase, preferably chromatography resin (ml swollen gel).

[0021] In a second aspect the invention relates to a split intein system comprising a N-intein as described above and a C-intein sequence which is co-expressed with a POI (protein of interest). The C-intein acts as a tag on the POI for binding to the N-intein attached to solid phase. After binding, the POI is cleaved off from the combined N-intein and C-intein and delivering a tagless POI. The C-intein variant is a split intein C-intein sequence or engineered variants thereof. A preferred C-intein sequence is mentioned in WO2021/099607 A1.

[0022] The POI's may be any recombinant proteins: proteins requiring native or near native N-terminal sequences, for example therapeutic protein candidates, biologics, antibody fragments, antibody mimetics, enzymes, recombinant proteins or peptides, such as growth factors, cytokines, chemokines, hormones, antigen (viral, bacterial, yeast, mammalian) production, vaccine production, cell surface receptors, fusion proteins.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] FIG. 1 shows a SDS-PAGE analysis of supernatants from different constructs after extraction using different techniques.

[0024] FIG. 2 shows solubility of different constructs determined after densitometric evaluation of SDS-PAGE analysis. Extracts from three different cell-cultures for each construct were analysed. Bars show the average solubility compared with whole cell lysate, (SDS) and the error bars show the standard deviation.

[0025] FIG. 3 shows N-intein concentrations in supernatants from different extracts determined by Biacore CFCA analysis. Extracts from three different cell-cultures for each construct were analysed. Bars show the average concentration and the error bars show the standard deviation.

[0026] FIG. 4 shows the ratio of N-intein in supernatants from extracts of different constructs using different extraction methods, compared as % to total amount of N-intein solubilized by SDS and heating.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

[0027] As used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a functional group,” “an alkyl,” or “a residue” includes mixtures of two or more such functional groups, alkyls, or residues, and the like.

[0028] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, a further aspect includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms a further aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint. It is also understood that there are a number of values disclosed herein, and that each value is also herein disclosed as “about” that particular value in addition to the value itself.

[0029] A weight percent (wt. %) of a component, unless specifically stated to the contrary, is based on the total weight of the formulation or composition in which the component is included.

[0030] As used herein, the terms “optional” or “optionally” means that the subsequently described event or circumstance can or can not occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

[0031] The term “contacting” as used herein refers to bringing two biological entities together in such a manner that the compound can affect the activity of the target, either directly; i.e., by interacting with the target itself, or indirectly; i.e., by interacting with another molecule, co-factor, factor, or protein on which the activity of the target is dependent. “Contacting” can also mean facilitating the interaction of two biological entities, such as peptides, to bond covalently or otherwise.

[0032] The term “peptide”, “polypeptides” and “protein” are used interchangeably herein and include proteins and fragments thereof. Polypeptides are disclosed herein as amino acid residue sequences. Those sequences are written left to right in the direction from the amino to the carboxy terminus. In accordance with standard nomenclature, amino acid residue sequences are denominated by either a three letter or a single letter code as indicated as follows: Alanine (Ala, A), Arginine (Arg, R), Asparagine (Asn, N), Aspartic Acid (Asp, D), Cysteine (Cys, C), Glutamine (Gln, Q), Glutamic Acid (Glu, E), Glycine (Gly, G), Histidine (His, H), Isoleucine (Ile, I), Leucine (Leu, L), Lysine (Lys, K), Methionine (Met, M), Phenylalanine (Phe, F), Proline (Pro, P), Serine (Ser, S), Threonine (Thr, T), Tryptophan (Trp, W), Tyrosine (Tyr, Y), and Valine (Val, V). Peptides include any oligopeptide, polypeptide, gene product, expression product, or protein. A peptide is comprised of consecutive amino acids and encompasses naturally occurring or synthetic molecules.

[0033] In addition, as used herein, the term “peptide” refers to amino acids joined to each other by peptide bonds or modified peptide bonds, e.g., peptide isosteres, etc. and may contain modified amino acids other than the 20 gene-encoded amino acids. The peptides can be modified by either natural processes, such as post-translational processing, or by chemical modification techniques which are well known in the art. Modifications can occur anywhere in the peptide, including the peptide backbone, the amino acid side-chains and the amino or carboxyl termini. The same type of modification can be present in the same or varying degrees at several sites in a given polypeptide. Also, a given peptide can have many types of modifications. Modifications include, without limitation, linkage of distinct domains or motifs, acetylation, acylation, ADP-ribosylation, amidation, covalent cross-linking or cyclization, covalent attachment of flavin, covalent attachment of a heme moiety, covalent attachment of a nucleotide or nucleotide derivative, covalent attachment of a lipid or lipid derivative, covalent attachment of a phosphatidylinositol, disulfide bond formation, demethylation, formation of cysteine or pyroglutamate, formylation, gamma-carboxylation, glycosylation, GPI anchor formation, hydroxylation, iodination, methylation, myristoylation, oxidation, pergylation, proteolytic processing, phosphorylation, prenylation, racemization, selenoylation, sulfation, and transfer-RNA mediated addition of amino acids to

protein such as arginylation. (See Proteins—Structure and Molecular Properties 2nd Ed., T. E. Creighton, W. H. Freeman and Company, New York (1993); Posttranslational Covalent Modification of Proteins, B. C. Johnson, Ed., Academic Press, New York, pp. 1-12 (1983)).

[0034] As used herein, “variant” refers to a molecule that retains a biological activity that is the same or substantially similar to that of the original sequence. The variant may be from the same or different species or be a synthetic sequence based on a natural or prior molecule. Moreover, as used herein, “variant” refers to a molecule having a structure attained from the structure of a parent molecule (e.g., a protein or peptide disclosed herein) and whose structure or sequence is sufficiently similar to those disclosed herein that based upon that similarity, would be expected by one skilled in the art to exhibit the same or similar activities and utilities compared to the parent molecule. For example, substituting specific amino acids in a given peptide can yield a variant peptide with similar activity to the parent.

[0035] In the context of the present invention, a substitution in a variant protein is indicated as: [original amino acid/position in sequence/substituted amino acid].

[0036] As used herein, the term “protein of interest (POI)” includes any synthetic or naturally occurring protein or peptide. The term therefore encompasses those compounds traditionally regarded as drugs, vaccines, and biopharmaceuticals including molecules such as proteins, peptides, and the like. Examples of therapeutic agents are described in well-known literature references such as the Merck Index (14th edition), the Physicians' Desk Reference (64th edition), and The Pharmacological Basis of Therapeutics (1st edition), and they include, without limitation, medicaments; substances used for the treatment, prevention, diagnosis, cure or mitigation of a disease or illness; substances that affect the structure or function of the body, or pro-drugs, which become biologically active or more active after they have been placed in a physiological environment.

[0037] As used herein, “isolated peptide” or “purified peptide” is meant to mean a peptide (or a fragment thereof) that is substantially free from the materials with which the peptide is normally associated in nature, or from the materials with which the peptide is associated in an artificial expression or production system, including but not limited to an expression host cell lysate, growth medium components, buffer components, cell culture supernatant, or components of a synthetic in vitro translation system. The peptides disclosed herein, or fragments thereof, can be obtained, for example, by extraction from a natural source (for example, a mammalian cell), by expression of a recombinant nucleic acid encoding the peptide (for example, in a cell or in a cell-free translation system), or by chemically synthesizing the peptide. In addition, peptide fragments may be obtained by any of these methods, or by cleaving full length proteins and/or peptides.

[0038] The word “or” as used herein means any one member of a particular list and also includes any combination of members of that list.

[0039] The phrase “nucleic acid” as used herein refers to a naturally occurring or synthetic oligonucleotide or polynucleotide, whether DNA or RNA or DNA-RNA hybrid, single-stranded or double-stranded, sense or antisense, which is capable of hybridization to a complementary nucleic acid by Watson-Crick base-pairing. Nucleic acids of the invention can also include nucleotide analogs (e.g., BrdU), and non-phosphodiester internucleoside linkages (e.g., peptide nucleic acid (PNA) or thiodiester linkages). In particular, nucleic acids can include, without limitation, DNA, RNA, cDNA, gDNA, ssDNA, dsDNA or any combination thereof.

[0040] As used herein, “extein” refers to the portion of an intein-modified protein that is not part of the intein and which can be spliced or cleaved upon excision of the intein.

[0041] “Intein” refers to an in-frame intervening sequence in a protein. An intein can catalyze its own excision from the protein through a post-translational protein splicing process to yield the free intein and a mature protein. An intein can also catalyze the cleavage of the intein-extein bond at either the intein N-terminus, or the intein C-terminus, or both of the intein-extein termini. As used herein, “intein” encompasses mini-inteins, modified or mutated inteins, and split inteins.

[0042] As used herein, the term “split intein” refers to any intein in which one or more peptide bond breaks exists between the N-terminal intein segment and the C-terminal intein segment such that the N-terminal and C-terminal intein segments become separate molecules that can non-covalently reassociate, or reconstitute, into an intein that is functional for splicing or cleaving reactions. Any catalytically active intein, or fragment thereof, may be used to derive a split intein for use in the systems and methods disclosed herein. For example, in one aspect the split intein may be derived from a eukaryotic intein. In another aspect, the split intein may be derived from a bacterial intein. In another aspect, the split intein may be derived from an archaeal intein. Preferably, the split intein so-derived will possess only the amino acid sequences essential for catalyzing splicing reactions.

[0043] As used herein, the “N-terminal intein segment” or “N-intein” refers to any intein sequence that comprises an N-terminal amino acid sequence that is functional for splicing and/or cleaving reactions when combined with a corresponding C-terminal intein segment. An N-terminal intein segment thus also comprises a sequence that is spliced out when splicing occurs. An N-terminal intein segment can comprise a sequence that is a modification of the N-terminal portion of a naturally occurring (native) intein sequence. Non-intein residues can also be genetically fused to intein segments to provide additional functionality, such as the ability to be affinity purified or to be covalently immobilized.

[0044] As used herein, the “C-terminal intein segment” or “C-intein” refers to any intein sequence that comprises a C-terminal amino acid sequence that is functional for splicing or cleaving reactions when combined with a corresponding N-terminal intein segment. In one aspect, the C-terminal intein segment comprises a sequence that is spliced out when splicing occurs. In another aspect, the C-terminal intein segment is cleaved from a peptide sequence fused to its C-terminus. The sequence which is cleaved from the C-terminal intein's C-terminus is referred to herein as a “protein of interest POI” is discussed in more detail below. A C-terminal intein segment can comprise a sequence that is a modification of the C-terminal portion of a naturally occurring (native) intein sequence. For example, a C terminal intein segment can comprise additional amino acid residues and/or mutated residues so long as the inclusion of such additional and/or mutated residues does not render the C-terminal intein segment non-functional for splicing or cleaving.

[0045] A consensus sequence is a sequence of DNA, RNA, or protein that represents aligned, related sequences. The consensus sequence of the related sequences can be defined in different ways, but is normally defined by the most common nucleotide(s) or amino acid residue(s) at each position.

[0046] As used herein, the term “splice” or “splices” means to excise a central portion of a polypeptide to form two or more smaller polypeptide molecules. In some cases, splicing also includes the step of fusing together two or more of the smaller polypeptides to form a new polypeptide. Splicing can also refer to the joining of two polypeptides encoded on two separate gene products through the action of a split intein.

[0047] As used herein, the term “cleave” or “cleaves” means to divide a single polypeptide to form two or more smaller polypeptide molecules. In some cases, cleavage is mediated by the addition of an extrinsic endopeptidase, which is often referred to as “proteolytic cleavage”. In other cases, cleaving can be mediated by the intrinsic activity of one or both of the cleaved peptide sequences, which is often referred to as “self-cleavage”. Cleavage can also refer to the self-cleavage of two polypeptides that is induced by the addition of a non-proteolytic third peptide, as in the action of split intein system described herein.

[0048] By the term “fused” is meant covalently bonded to. For example, a first peptide is fused to a second peptide when the two peptides are covalently bonded to each other (e.g., via a peptide bond).

[0049] As used herein an “isolated” or “substantially pure” substance is one that has been separated from components which naturally accompany it. Typically, a polypeptide is substantially pure when

it is at least 50% (e.g., 60%, 70%, 80%, 90%, 95%, and 99%) by weight free from the other proteins and naturally-occurring organic molecules with which it is naturally associated.

[0050] Herein, “bind” or “binds” means that one molecule recognizes and adheres to another molecule in a sample, but does not substantially recognize or adhere to other molecules in the sample. One molecule “specifically binds” another molecule if it has a binding affinity greater than about 10^{sup.5} to 10^{sup.6} liters/mole for the other molecule.

[0051] Nucleic acids, nucleotide sequences, proteins or amino acid sequences referred to herein can be isolated, purified, synthesized chemically, or produced through recombinant DNA technology. All of these methods are well known in the art.

[0052] As used herein, the terms “modified” or “mutated,” as in “modified intein” or “mutated intein,” refer to one or more modifications in either the nucleic acid or amino acid sequence being referred to, such as an intein, when compared to the native, or naturally occurring structure. Such modification can be a substitution, addition, or deletion. The modification can occur in one or more amino acid residues or one or more nucleotides of the structure being referred to, such as an intein.

[0053] As used herein, the term “modified peptide”, “modified protein” or “modified protein of interest” or “modified target protein” refers to a protein which has been modified.

[0054] As used herein, “operably linked” refers to the association of two or more biomolecules in a configuration relative to one another such that the normal function of the biomolecules can be performed. In relation to nucleotide sequences, “operably linked” refers to the association of two or more nucleic acid sequences, by means of enzymatic ligation or otherwise, in a configuration relative to one another such that the normal function of the sequences can be performed. For example, the nucleotide sequence encoding a pre-sequence or secretory leader is operably linked to a nucleotide sequence for a polypeptide if it is expressed as a pre-protein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the coding sequence; and a ribosome binding site is operably linked to a coding sequence if it is positioned so as to facilitate translation of the sequence.

[0055] “Sequence homology” can refer to the situation where nucleic acid or protein sequences are similar because they have a common evolutionary origin. “Sequence homology” can indicate that sequences are very similar. Sequence similarity is observable; homology can be based on the observation. “Very similar” can mean at least 70% identity, homology or similarity; at least 75% identity, homology or similarity; at least 80% identity, homology or similarity; at least 85% identity, homology or similarity; at least 90% identity, homology or similarity; such as at least 93% or at least 95% or even at least 97% identity, homology or similarity. The nucleotide sequence similarity or homology or identity can be determined using the “Align” program of Myers et al. (1988) CABIOS 4:11-17 and available at NCBI. Additionally or alternatively, amino acid sequence similarity or identity or homology can be determined using the BlastP program (Altschul et al. Nucl. Acids Res. 25:3389-3402), and available at NCBI. Alternatively or additionally, the terms “similarity” or “identity” or “homology,” for instance, with respect to a nucleotide sequence, are intended to indicate a quantitative measure of homology between two sequences.

[0056] Alternatively or additionally, “similarity” with respect to sequences refers to the number of positions with identical nucleotides divided by the number of nucleotides in the shorter of the two sequences wherein alignment of the two sequences can be determined in accordance with the Wilbur and Lipman algorithm. (1983) Proc. Natl. Acad. Sci. USA 80:726. For example, using a window size of 20 nucleotides, a word length of 4 nucleotides, and a gap penalty of 4, and computer-assisted analysis and interpretation of the sequence data including alignment can be conveniently performed using commercially available programs (e.g., Intelligenetics™ Suite, Intelligenetics™ Inc. CA). When RNA sequences are said to be similar, or have a degree of sequence identity with DNA sequences, thymidine (T) in the DNA sequence is considered equal to uracil (U) in the RNA sequence. The following references also provide algorithms for comparing the relative identity or homology or similarity of amino acid residues of two proteins, and

additionally or alternatively with respect to the foregoing, the references can be used for determining percent homology or identity or similarity. Needleman et al. (1970) J. Mol. Biol. 48:444-453; Smith et al. (1983) Advances App. Math. 2:482-489; Smith et al. (1981) Nuc. Acids Res. 11:2205-2220; Feng et al. (1987) J. Molec. Evol. 25:351-360; Higgins et al. (1989) CABIOS 5:151-153; Thompson et al. (1994) Nuc. Acids Res. 22:4673-480; and Devereux et al. (1984) 12:387-395. "Stringent hybridization conditions" is a term which is well known in the art; see, for example, Sambrook, "Molecular Cloning, A Laboratory Manual" second ed., CSH Press, Cold Spring Harbor, 1989; "Nucleic Acid Hybridization, A Practical Approach", Hames and Higgins eds., IRL Press, Oxford, 1985; see also FIG. 2 and description thereof herein wherein there is a sequence comparison.

[0057] The term "buffer" or "buffered solution" refers to solutions which resist changes in pH by the action of its conjugate acid-base range.

[0058] The term "loading buffer" or "equilibrium buffer" refers to the buffer containing the salt or salts which is mixed with the protein preparation for loading the protein preparation onto a column. This buffer is also used to equilibrate the column before loading, and to wash to column after loading the protein.

[0059] The term "wash buffer" is used herein to refer to the buffer that is passed over a column (for example) following loading of a protein of interest (such as one coupled to a C-terminal intein fragment, for example) and prior to elution of the protein of interest. The wash buffer may serve to remove one or more contaminants without substantial elution of the desired protein.

[0060] The term "elution buffer" refers to the buffer used to elute the desired protein from the column. As used herein, the term "solution" refers to either a buffered or a non-buffered solution, including water.

[0061] The term "washing" means passing an appropriate buffer through or over a solid support, such as a chromatographic resin.

[0062] The term "eluting" a molecule (e.g. a desired protein or contaminant) from a solid support means removing the molecule from such material.

[0063] The term "contaminant" or "impurity" refers to any foreign or objectionable molecule, particularly a biological macromolecule such as a DNA, an RNA, or a protein, other than the protein being purified, that is present in a sample of a protein being purified. Contaminants include, for example, other proteins from cells that express and/or secrete the protein being purified.

[0064] The term "separate" or "isolate" as used in connection with protein purification refers to the separation of a desired protein from a second protein or other contaminant or mixture of impurities in a mixture comprising both the desired protein and a second protein or other contaminant or impurity mixture, such that at least the majority of the molecules of the desired protein are removed from that portion of the mixture that comprises at least the majority of the molecules of the second protein or other contaminant or mixture of impurities.

[0065] The term "purify" or "purifying" a desired protein from a composition or solution comprising the desired protein and one or more contaminants means increasing the degree of purity of the desired protein in the composition or solution by removing (completely or partially) at least one contaminant from the composition or solution.

N-Intein Protein Variants

[0066] The invention relates to affinity chromatography and affinity tag cleavage mechanisms in a single step using a split intein system according to the invention which cleaves with broad amino acid tolerance to generate a tag less protein of interest (POI) as end product. The two halves of the intein are the affinity ligand (N-intein) and the affinity tag (C-intein) and they associate rapidly. Immobilizing one half (N-intein) on a chromatography resin enables the capture of the other half (C-intein) coupled to the POI from solution. In the presence of Zn^{sup.2+} ions, the cleavage reaction is inhibited, enabling a stable complex to form while impurities are washed away. After impurities are eliminated, a chelator or reducing agent is added, and the cleavage reaction proceeds,

enabling collection of the POI, while the intein tag remains bound non-covalently to the cognate intein linked to the chromatography resin.

[0067] Preferably the invention provides N-intein protein variant sequences of native split inteins or consensus sequences derived from native inteins and split inteins wherein, the N-intein variant is modified as compared to the native sequence or consensus sequence to provide increased solubility by having mutations in position 24 and/or 25. These positions are calculated according to conventional clustal alignment with native split inteins starting from the initial catalytical cysteine which is number 1.

[0068] Native intein are known in the art. A list of inteins is found in Table 1 below. All inteins have the potential to be made into split inteins while some inteins naturally exist in split form. All of the inteins found in the table either exist as split inteins or have the potential to be made into split inteins modified in accordance with the invention at position 24 and/or 25 such that the increased solubility is achieved compared to the native sequences.

TABLE-US-00001 TABLE 1 Naturally Occurring Inteins

Intein Name	Organism Name	Organism Description
APMV Pol	<i>Acanthamoeba polyphaga</i>	Mimivirus isolate = "Rowbotham-Bradford", Virus, infects Amoebae, taxon: 212035
PRP8	<i>Aspergillus brevipes</i>	FRR2439 Fungi, ATCC 16899, taxon: 75551
Aca-G186AR	<i>Ajellomyces capsulatus</i>	G186AR Taxon: 447093, strain G186AR
Aca-H143	<i>Ajellomyces capsulatus</i>	H143 Taxon: 544712
Aca-JER2004	<i>Ajellomyces capsulatus</i>	(anamorph: strain = JER2004, taxon: 5037, <i>Histoplasma capsulatum</i>)
PRP8	<i>Ajellomyces capsulatus</i>	NAm1 strain = "NAm1", taxon: 339724
Ade-ER3	<i>Ajellomyces dermatitidis</i>	ER-3 Human fungal pathogen, taxon: 559297
Ade-SLH14081	<i>Ajellomyces dermatitidis</i>	SLH14081, Human fungal pathogen
Afu-Af293	<i>Aspergillus fumigatus</i>	var. <i>ellipticus</i> , Human pathogenic fungus, strain Af293 taxon: 330879
Afu-FRR0163	<i>Aspergillus fumigatus</i>	strain Human pathogenic fungus, FRR0163 taxon: 5085
NRRL5109	<i>Aspergillus fumigatus</i>	var. <i>ellipticus</i> , Human pathogenic fungus, strain NRRL 5109 taxon: 41121
Ag-NRRL6136	<i>Aspergillus giganteus</i>	Strain NRRL Fungus, taxon: 5060
FGSC A	<i>Aspergillus nidulans</i>	Filamentous fungus, taxon: 227321
Avi	<i>Aspergillus viridinutans</i>	strain Fungi, ATCC 16902, FRR0577 taxon: 75553
Bci	<i>Botrytis cinerea</i>	(teleomorph of Plant fungal pathogen <i>Botryotinia fuckeliana</i> B05.10)
Bde-JEL197	<i>Batrachochytrium dendrobatidis</i>	Chytrid fungus, JEL197 isolate = "AFTOL-ID 21", taxon: 109871
Bde-JEL423	<i>Batrachochytrium dendrobatidis</i>	Chytrid fungus, isolate JEL423 JEL423, taxon 403673
Bde-JEL423	<i>Batrachochytrium dendrobatidis</i>	Chytrid fungus, isolate JEL423 JEL423, taxon 403673
Bde-JEL423	<i>Batrachochytrium dendrobatidis</i>	Chytrid fungus, isolate JEL423 JEL423, taxon 403673
Bde-JEL423	<i>Batrachochytrium dendrobatidis</i>	Chytrid fungus, isolate JEL423 JEL423, taxon 403673
Bfu-B05	<i>Botryotinia fuckeliana</i>	B05.10 Taxon: 332648
CIV RIR1	Chilo iridescent virus	dsDNA eucaryotic virus, taxon: 10488
CV-NY2A	<i>Chlorella</i> virus NY2A	infects dsDNA eucaryotic <i>Chlorella</i> NC64A, which infects virus, taxon: 46021, Family <i>Paramecium bursaria</i>
Phycodnaviridae	<i>Chlorella</i> virus NY2A	infects dsDNA eucaryotic <i>Chlorella</i> NC64A, which infects virus, taxon: 46021, Family <i>Paramecium bursaria</i>
CZIV RIR1	<i>Costelytra zealandica</i>	iridescent virus dsDNA eucaryotic virus, Taxon: 68348
Cba-WM02.98	<i>Cryptococcus bacillisporus</i>	strain Yeast, human pathogen, WM02.98 (aka <i>Cryptococcus</i> taxon: 37769
Cba-WM728	<i>Cryptococcus bacillisporus</i>	strain Yeast, human pathogen, WM728 taxon: 37769
Ceu ClpP	<i>Chlamydomonas eugametos</i>	Green alga, taxon: 3053 (chloroplast)
Cga	<i>Cryptococcus gattii</i>	(aka Yeast, human pathogen <i>Cryptococcus bacillisporus</i>)
Cgl VMA	<i>Candida glabrata</i>	Yeast, taxon: 5478
Cla	<i>Cryptococcus laurentii</i>	strain Fungi, Basidiomycete yeast, CBS139 taxon: 5418
Cmo ClpP	<i>Chlamydomonas moewusii</i>	strain Green alga, chloroplast gene, UTEX 97 taxon: 3054
Cmo RPB2 (RpoBb)	<i>Chlamydomonas moewusii</i>	strain Green alga, chloroplast gene, UTEX 97 taxon: 3054
Cne-A	<i>Filobasidiella neoformans</i>	Yeast, human pathogen PRP8) (<i>Cryptococcus neoformans</i>) Serotype A, PHLS_8104
Cne-AD	<i>Filobasidiella neoformans</i>	Yeast, human pathogen PRP8) (<i>Cryptococcus neoformans</i>) Serotype A, PHLS_8104

Cryptococcus neoformans Yeast, human pathogen, AD PRP8) (*Filobasidiella neoformans*), ATCC32045, taxon: 5207 Serotype AD, CBS132). Cne-JEC21 PRP8 *Cryptococcus neoformans* var. Yeast, human pathogen, *neoformans* JEC21 serotype = “D” taxon: 214684 Cpa ThrRS *Candida parapsilosis*, strain CLIB214 Yeast, Fungus, taxon: 5480 Cre RPB2 *Chlamydomonas reinhardtii* Green algae, taxon: 3055 (nucleus) CroV Pol *Cafeteria roenbergensis* virus BV- taxon: 693272, Giant virus PW1 infecting marine heterotrophic nanoflagellate CroV RIR1 *Cafeteria roenbergensis* virus BV- taxon: 693272, Giant virus PW1 infecting marine heterotrophic nanoflagellate CroV RPB2 *Cafeteria roenbergensis* virus BV- taxon: 693272, Giant virus PW1 infecting marine heterotrophic nanoflagellate CroV Top2 *Cafeteria roenbergensis* virus BV- taxon: 693272, Giant virus PW1 infecting marine heterotrophic nanoflagellate Cst RPB2 *Coelomomyces stegomyiae* Chytrid fungus, isolate = “AFTOL-ID 18”, taxon: 143960 Ctr ThrRS *Candida tropicalis* ATCC750 Yeast Ctr VMA *Candida tropicalis* (nucleus) Yeast Ctr-MYA3404 VMA *Candida tropicalis* MYA-3404 Taxon: 294747 Ddi RPC2 *Dictyostelium discoideum* strain Mycetoza (a social amoeba) AX4 (nucleus) Dhan GLT1 *Debaryomyces hansenii* CBS767 Fungi, Anamorph: *Candida famata*, taxon: 4959 Dhan VMA *Debaryomyces hansenii* CBS767 Fungi, taxon: 284592 Eni PRP8 *Emericella nidulans* R20 (anamorph: taxon: 162425 *Aspergillus nidulans*) Eni-FGSCA4 PRP8 *Emericella nidulans* (anamorph: Filamentous fungus, *Aspergillus nidulans*) FGSC A4 taxon: 162425 Fte RPB2 (RpoB) *Floydiella terrestris*, strain UTEX Green alga, chloroplast gene, 1709 taxon: 51328 Gth DnaB *Guillardia theta* (plastid) Cryptophyte Algae HaV01 Pol *Heterosigma akashiwo* virus 01 Algal virus, taxon: 97195, strain HaV01 Hca PRP8 *Histoplasma capsulatum* (anamorph: Fungi, human pathogen *Ajellomyces capsulatus*) IIV6 RIR1 Invertebrate iridescent virus 6 dsDNA eucaryotic virus, taxon: 176652 Kex-CBS379 VMA *Kazachstania exigua*, formerly Yeast, taxon: 34358 *Saccharomyces exiguus*, strain CBS379 Kla-CBS683 VMA *Kluyveromyces lactis*, strain CBS683 Yeast, taxon: 28985 Kla-IFO1267 VMA *Kluyveromyces lactis* IFO1267 Fungi, taxon: 28985 Kla-NRRLY1140 VMA *Kluyveromyces lactis* NRRL Y-1140 Fungi, taxon: 284590 Lel VMA *Lodderomyces elongisporus* Yeast Mca-CBS113480 PRP8 *Microsporum canis* CBS 113480 Taxon: 554155 Nau PRP8 *Neosartorya aurata* NRRL 4378 Fungus, taxon: 41051 Nfe-NRRL5534 PRP8 *Neosartorya fennelliae* NRRL 5534 Fungus, taxon: 41048 Nfi PRP8 *Neosartorya fischeri* Fungi Ngl-FR2163 PRP8 *Neosartorya glabra* FRR2163 Fungi, ATCC 16909, taxon: 41049 Ngl-FRR1833 PRP8 *Neosartorya glabra* FRR1833 Fungi, taxon: 41049, (preliminary identification) Nqu PRP8 *Neosartorya quadricincta*, strain taxon: 41053 NRRL 4175 Nspi PRP8 *Neosartorya spinosa* FRR4595 Fungi, taxon: 36631 Pabr-Pb01 PRP8 *Paracoccidioides brasiliensis* Pb01 Taxon: 502779 Pabr-Pb03 PRP8 *Paracoccidioides brasiliensis* Pb03 Taxon: 482561 Pan CHS2 *Podospira anserina* Fungi, Taxon 5145 Pan GLT1 *Podospira anserina* Fungi, Taxon 5145 Pbl PRP8-a *Phycomyces blakesleeanus* Zygomycete fungus, strain NRRL155 Pbl PRP8-b *Phycomyces blakesleeanus* Zygomycete fungus, strain NRRL155 Pbr-Pb18 PRP8 *Paracoccidioides brasiliensis* Pb18 Fungi, taxon: 121759 Pch PRP8 *Penicillium chrysogenum* Fungus, taxon: 5076 Pex PRP8 *Penicillium expansum* Fungus, taxon: 27334 Pgu GLT1 *Pichia (Candida) guilliermondii* Fungi, Taxon 294746 Pgu-alt GLT1 *Pichia (Candida) guilliermondii* Fungi Pno GLT1 *Phaeosphaeria nodorum* SN15 Fungi, taxon: 321614 Pno RPA2 *Phaeosphaeria nodorum* SN15 Fungi, taxon: 321614 Ppu DnaB *Porphyra purpurea* (chloroplast) Red Alga Pst VMA *Pichia stipitis* CBS 6054, Yeast taxon: 322104 Ptr PRP8 *Pyrenophora tritici-repentis* Pt-1C- Ascomycete BF fungus, taxon: 426418 Pvu PRP8 *Penicillium vulpinum* (formerly Fungus *P. claviforme*) Pye DnaB *Porphyra yezoensis* chloroplast, Red alga, cultivar U-51 organelle = “plastid: chloroplast”, “taxon: 2788 Sas RPB2 *Spiromyces spiralis* NRRL 22631 Zygomycete fungus, isolate = “AFTOL-ID 185”, taxon: 68401 Sca-CBS4309 VMA *Saccharomyces castellii*, strain Yeast, taxon: 27288 CBS4309 Sca-IFO1992 VMA *Saccharomyces castellii*, strain Yeast, taxon: 27288 IFO1992 Scar VMA *Saccharomyces cariocanus*, Yeast, taxon: 114526 strain = “UFRJ 50791 Sce VMA *Saccharomyces cerevisiae* (nucleus) Yeast, also in Sce strains OUT7163, OUT7045, OUT7163, IFO1992 Sce-DH1-1A VMA *Saccharomyces cerevisiae* strain Yeast, taxon: 173900, also in DH1-1A Sce strains

OUT7900, OUT7903, OUT7112 Sce-JAY291 VMA *Saccharomyces cerevisiae* JAY291 Taxon: 574961 Sce-OUT7091 VMA *Saccharomyces cerevisiae* OUT7091 Yeast, taxon: 4932, also in Sce strains OUT7043, OUT7064 Sce-OUT7112 VMA *Saccharomyces cerevisiae* OUT7112 Yeast, taxon: 4932, also in Sce strains OUT7900, OUT7903 Sce-YJM789 VMA *Saccharomyces cerevisiae* strain Yeast, taxon: 307796 YJM789 Sda VMA *Saccharomyces dairenensis*, strain Yeast, taxon: 27289, Also in CBS 421 Sda strain IFO0211 Sex-IFO1128 VMA *Saccharomyces exiguus*, Yeast, taxon: 34358 strain = "IFO1128" She RPB2 (RpoB) *Stigeoclonium helveticum*, strain Green alga, chloroplast gene, UTEX 441 taxon: 55999 Sja VMA *Schizosaccharomyces japonicus* Ascomycete fungus, yFS275 taxon: 402676 Spa VMA *Saccharomyces pastorianus* Yeast, taxon: 27292 IFO11023 Spu PRP8 *Spizellomyces punctatus* Chytrid fungus, Sun VMA *Saccharomyces unisporus*, strain Yeast, taxon: 27294 CBS 398 Tgl VMA *Torulaspora globosa*, strain CBS 764 Yeast, taxon: 48254 Tpr VMA *Torulaspora pretoriensis*, strain CBS Yeast, taxon: 35629 5080 Ure-1704 PRP8 *Uncinocarpus reesii* Filamentous fungus Vpo VMA *Vanderwaltozyma polyspora*, Yeast, taxon: 36033 formerly *Kluyveromyces polysporus*, strain CBS 2163 WIV RIR1 *Wiseana* iridescent virus dsDNA eucaryotic virus, taxon: 68347 Zba VMA *Zygosaccharomyces bailii*, strain Yeast, taxon: 4954 CBS 685 Zbi VMA *Zygosaccharomyces bisporus*, strain Yeast, taxon: 4957 CBS 702 Zro VMA *Zygosaccharomyces rouxii*, strain Yeast, taxon: 4956 CBS 688 Eubacteria AP-APSE1 dpol *Acyrrhosiphon pisum* secondary Bacteriophage, taxon: 67571 endosymbiot phage 1 AP-APSE2 dpol Bacteriophage APSE-2, isolate = T5A Bacteriophage of *Candidatus Hamiltonella defensa*, endosymbiot of *Acyrrhosiphon pisum*, taxon: 340054 AP-APSE4 dpol Bacteriophage of *Candidatus* Bacteriophage, taxon: 568990 *Hamiltonella defensa* strain 5ATac, endosymbiot of *Acyrrhosiphon pisum* AP-APSE5 dpol Bacteriophage APSE-5 Bacteriophage of *Candidatus Hamiltonella defensa*, endosymbiot of *Uroleucon rudbeckiae*, taxon: 568991 AP-Aaphi23 MupF Bacteriophage Aaphi23, *Actinobacillus Haemophilus* phage Aaphi23 *actinomycetemcomitans* Bacteriophage, taxon: 230158 Aae RIR2 *Aquifex aeolicus* strain VF5 Thermophilic chemolithoautotroph, taxon: 63363 Aave-AAC001 *Acidovorax avenae* subsp. *citrulli* taxon: 397945 Aave1721 AAC00-1 Aave-AAC001 RIR1 *Acidovorax avenae* subsp. *citrulli* taxon: 397945 AAC00-1 Aave-ATCC19860 *Acidovorax avenae* subsp. *avenae* Taxon: 643561 RIR1 ATCC 19860 Aba Hyp-02185 *Acinetobacter baumannii* ACICU taxon: 405416 Ace RIR1 *Acidothermus cellulolyticus* 11B taxon: 351607 Aeh DnaB-1 *Alkalilimnicola ehrlichei* MLHE-1 taxon: 187272 Aeh DnaB-2 *Alkalilimnicola ehrlichei* MLHE-1 taxon: 187272 Aeh RIR1 *Alkalilimnicola ehrlichei* MLHE-1 taxon: 187272 AgP-S1249 MupF *Aggregatibacter* phage S1249 Taxon: 683735 Aha DnaE-c *Aphanothece halophytica* *Cyanobacterium*, taxon: 72020 Aha DnaE-n *Aphanothece halophytica* *Cyanobacterium*, taxon: 72020 Alvi-DSM180 GyrA *Allochromatium vinosum* DSM 180 Taxon: 572477 Ama MADE823 phage uncharacterized protein Probably prophage gene, [*Alteromonas macleodii* 'Deep taxon: 314275 ecotype'] Amax-CS328 DnaX *Arthrospira maxima* CS-328 Taxon: 513049 Aov DnaE-c *Aphanizomenon ovalisporum* *Cyanobacterium*, taxon: 75695 Aov DnaE-n *Aphanizomenon ovalisporum* *Cyanobacterium*, taxon: 75695 Apl-C1 DnaX *Arthrospira platensis* Taxon: 118562, strain C1 Arsp-FB24 DnaB *Arthrobacter* species FB24 taxon: 290399 Asp DnaE-c *Anabaena* species PCC7120, (*Nostoc* *Cyanobacterium*, Nitrogen- sp. PCC7120) fixing, taxon: 103690 Asp DnaE-n *Anabaena* species PCC7120, (*Nostoc* *Cyanobacterium*, Nitrogen- sp. PCC7120) fixing, taxon: 103690 Ava DnaE-c *Anabaena variabilis* ATCC29413 *Cyanobacterium*, taxon: 240292 Ava DnaE-n *Anabaena variabilis* ATCC29413 *Cyanobacterium*, taxon: 240292 Avin RIR1 BIL *Azotobacter vinelandii* taxon: 354 Bce-MCO3 DnaB *Burkholderia cenocepacia* MC0-3 taxon: 406425 Bce-PC184 DnaB *Burkholderia cenocepacia* PC184 taxon: 350702 Bse-MLS10 TerA *Bacillus selenitireducens* MLS10 Probably prophage gene, Taxon: 439292 BsuP-M1918 RIR1 *B. subtilis* M1918 (prophage) Prophage in *B. subtilis* M1918. taxon: 157928 BsuP-SPBc2 RIR1 *B. subtilis* strain 168 Sp beta c2 *B. subtilis* taxon 1423. SPbeta prophage c2 phage, taxon: 66797 Bvi IcmO *Burkholderia vietnamiensis* G4 plasmid = "pBVIE03". taxon: 269482 CP-P1201 Thy1 *Corynebacterium* phage P1201 lytic bacteriophage

P1201 from *Corynebacterium glutamicum* NCHU 87078. Viruses; dsDNA viruses, taxon: 384848
 Cag RIR1 *Chlorochromatium aggregatum* Motile, phototrophic consortia Cau SpoVR *Chloroflexus aurantiacus* J-10-fl Anoxygenic phototroph, taxon: 324602 CbP-C-St RNR *Clostridium botulinum* phage C-St Phage, specific_host = "*Clostridium botulinum* type C strain C-Stockholm, taxon: 12336 CbP-D1873 RNR *Clostridium botulinum* phage D Ssp. phage from *Clostridium botulinum* type D strain, 1873, taxon: 29342 Cbu-Dugway DnaB *Coxiella burnetii* Dugway 5J108-111 Proteobacteria; Legionellales; taxon: 434922 Cbu-Goat DnaB *Coxiella burnetii* 'MSU Goat Q177' Proteobacteria; Legionellales; taxon: 360116 Cbu-RSA334 DnaB *Coxiella burnetii* RSA 334 Proteobacteria; Legionellales; taxon: 360117 Cbu-RSA493 DnaB *Coxiella burnetii* RSA 493 Proteobacteria; Legionellales; taxon: 227377 Cce Hyp1-Csp-2 *Cyanothece* sp. ATCC 51142 Marine unicellular diazotrophic cyanobacterium, taxon: 43989 Cch RIR1 *Chlorobium chlorochromatii* CaD3 taxon: 340177 Ccy Hyp1-Csp-1 *Cyanothece* sp. CCY0110 *Cyanobacterium*, taxon: 391612 Ccy Hyp1-Csp-2 *Cyanothece* sp. CCY0110 *Cyanobacterium*, taxon: 391612 Cfl-DSM20109 DnaB *Cellulomonas flavigena* DSM 20109 Taxon: 446466 Chy RIR1 *Carboxydotherrmus* Thermophile, taxon = 246194 *hydrogenoformans* Z-2901 Ckl PTerm *Clostridium kluyveri* DSM 555 plasmid = "pCKL555A", taxon: 431943 Cra-CS505 DnaE-c *Cylindrospermopsis raciborskii* CS-505 Taxon: 533240 Cra-CS505 DnaE-n *Cylindrospermopsis raciborskii* CS-505 Taxon: 533240 Cra-CS505 GyrB *Cylindrospermopsis raciborskii* CS-505 Taxon: 533240 Csp-CCY0110 DnaE-c *Cyanothece* sp. CCY0110 Taxon: 391612 Csp-CCY0110 DnaE-n *Cyanothece* sp. CCY0110 Taxon: 391612 Csp-PCC7424 DnaE-c *Cyanothece* sp. PCC 7424 *Cyanobacterium*, taxon: 65393 Csp-PCC7424 DnaE-n *Cyanothece* sp. PCC7424 *Cyanobacterium*, taxon: 65393 Csp-PCC7425 DnaB *Cyanothece* sp. PCC 7425 Taxon: 395961 Csp-PCC7822 DnaE-n *Cyanothece* sp. PCC 7822 Taxon: 497965 Csp-PCC8801 DnaE-c *Cyanothece* sp. PCC 8801 Taxon: 41431 Csp-PCC8801 DnaE-n *Cyanothece* sp. PCC 8801 Taxon: 41431 Cth ATPase BIL *Clostridium thermocellum* ATCC27405, taxon: 203119 Cth-ATCC27405 TerA *Clostridium thermocellum* Probable prophage, ATCC27405 ATCC27405, taxon: 203119 Cth-DSM2360 TerA *Clostridium thermocellum* DSM Probably prophage 2360 gene, Taxon: 572545 Cwa DnaB *Crocospaera watsonii* WH 8501 taxon: 165597 (*Synechocystis* sp. WH 8501) Cwa DnaE-c *Crocospaera watsonii* WH 8501 *Cyanobacterium*, (*Synechocystis* sp. WH 8501) taxon: 165597 Cwa DnaE-n *Crocospaera watsonii* WH 8501 *Cyanobacterium*, (*Synechocystis* sp. WH 8501) taxon: 165597 Cwa PEP *Crocospaera watsonii* WH 8501 taxon: 165597 (*Synechocystis* sp. WH 8501) Cwa RIR1 *Crocospaera watsonii* WH 8501 taxon: 165597 (*Synechocystis* sp. WH 8501) Daud RIR1 *Candidatus Desulforudis audaxviator* taxon: 477974 MP104C Dge DnaB *Deinococcus geothermalis* Thermophilic, radiation DSM11300 resistant Dha-DCB2 RIR1 *Desulfotobacterium hafniense* DCB-2 Anaerobic dehalogenating bacteria, taxon: 49338 Dha-Y51 RIR1 *Desulfotobacterium hafniense* Y51 Anaerobic dehalogenating bacteria, taxon: 138119 Dpr-MLMSl RIR1 delta proteobacterium MLMS-1 Taxon: 262489 Dra RIR1 *Deinococcus radiodurans* R1, TIGR Radiation resistant, strain taxon: 1299 Dra Snf2-c *Deinococcus radiodurans* R1, TIGR Radiation and DNA damage strain resistant, taxon: 1299 Dra Snf2-n *Deinococcus radiodurans* R1, TIGR Radiation and DNA damage strain resistant, taxon: 1299 Dra-ATCC13939 Snf2 *Deinococcus radiodurans* R1, Radiation and DNA damage ATCC13939/Brooks & Murray strain resistant, taxon: 1299 Dth UDP GD *Dictyoglomus thermophilum* H-6-12 strain = "H-6-12; ATCC 35947, taxon: 309799 Dvul ParB *Desulfovibrio vulgaris* subsp. taxon: 391774 *vulgaris* DP4 EP-Min27 Primase Enterobacteria phage Min27 bacteriophage of host = "*Escherichia coli* O157: H7 str. Min27" Fal DnaB *Frankia alni* ACN14a Plant symbiot, taxon: 326424 Fsp-CcI3 RIR1 *Frankia* species CcI3 taxon: 106370 Gob DnaE *Gemmata obscuriglobus* UQM2246 Taxon 114, TIGR genome strain, budding bacteria Gob Hyp *Gemmata obscuriglobus* UQM2246 Taxon 114, TIGR genome strain, budding bacteria Gvi DnaB *Gloeobacter violaceus*, PCC 7421 taxon: 33072 Gvi RIR1-1 *Gloeobacter violaceus*, PCC 7421 taxon: 33072 Gvi RIR1-2 *Gloeobacter violaceus*, PCC 7421 taxon: 33072 Hhal DnaB *Halorhodospira halophila* SL1 taxon: 349124 Kfl-DSM17836 DnaB *Kribbella flavida* DSM 17836

Taxon: 479435 Kra DnaB *Kineococcus radiotolerans* Radiation resistant SRS30216 LLP-KSY1
 PolA *Lactococcus* phage KSY1 Bacteriophage, taxon: 388452 LP-phiHSIC Helicase *Listonella pelagia* phage phiHSIC taxon: 310539, a pseudotemperate marine phage of *Listonella pelagia* Lsp-
 PCC8106 GyrB *Lyngbya* sp. PCC 8106 Taxon: 313612 MP-Be DnaB Mycobacteriophage
 Bethlehem Bacteriophage, taxon: 260121 MP-Be gp51 Mycobacteriophage Bethlehem
 Bacteriophage, taxon: 260121 MP-Catera gp206 Mycobacteriophage Catera Mycobacteriophage,
 taxon: 373404 MP-KBG gp53 *Mycobacterium* phage KBG Taxon: 540066 MP-McJw1 DnaB
 Mycobacteriophage CJW1 Bacteriophage, taxon: 205869 MP-Omega DnaB Mycobacteriophage
 Omega Bacteriophage, taxon: 205879 MP-U2 gp50 Mycobacteriophage U2 Bacteriophage, taxon:
 260120 Maer-NIES843 DnaB *Microcystis aeruginosa* NIES-843 Bloom-forming toxic
cyanobacterium, taxon: 449447 Maer-NIES843 DnaE-c *Microcystis aeruginosa* NIES-843 Bloom-
 forming toxic *cyanobacterium*, taxon: 449447 Maer-NIES843 DnaE-n *Microcystis aeruginosa*
 NIES-843 Bloom-forming toxic *cyanobacterium*, taxon: 449447 Mau-ATCC27029
Micromonospora aurantiaca ATCC Taxon: 644283 GyrA 27029 Mav-104 DnaB *Mycobacterium*
avium 104 taxon: 243243 Mav-ATCC25291 *Mycobacterium avium* subsp. *avium* Taxon: 553481
 DnaB ATCC 25291 Mav-ATCC35712 *Mycobacterium avium* ATCC35712, taxon 1764 DnaB Mav-
 PT DnaB *Mycobacterium avium* subsp. taxon: 262316 *paratuberculosis* str. k10 Mbo Pps1
Mycobacterium bovis subsp. *bovis* strain = "AF2122/97", AF2122/97 taxon: 233413 Mbo RecA
Mycobacterium bovis subsp. *bovis* taxon: 233413 AF2122/97 Mbo SufB (Mbo Pps1)
Mycobacterium bovis subsp. *bovis* taxon: 233413 AF2122/97 Mbo-1173P DnaB *Mycobacterium*
bovis BCG Pasteur strain = BCG Pasteur 1173P 1173P2,, taxon: 410289 Mbo-AF2122 DnaB
Mycobacterium bovis subsp. *bovis* strain = "AF2122/97", AF2122/97 taxon: 233413 Mca MupF
Methylococcus capsulatus Bath, prophage MuMc02, prophage MuMc02 taxon: 243233 Mca RIR1
Methylococcus capsulatus Bath taxon: 243233 Mch RecA *Mycobacterium chitae* IP14116003,
 taxon: 1792 Mcht-PCC7420 DnaE-1 *Microcoleus chthonoplastes* *Cyanobacterium*, PCC7420
 taxon: 118168 Mcht-PCC7420 DnaE-2c *Microcoleus chthonoplastes* *Cyanobacterium*, PCC7420
 taxon: 118168 Mcht-PCC7420 DnaE-2n *Microcoleus chthonoplastes* *Cyanobacterium*, PCC7420
 taxon: 118168 Mcht-PCC7420 GyrB *Microcoleus chthonoplastes* PCC Taxon: 118168 7420 Mcht-
 PCC7420 RIR1-1 *Microcoleus chthonoplastes* PCC Taxon: 118168 7420 Mcht-PCC7420 RIR1-2
Microcoleus chthonoplastes PCC Taxon: 118168 7420 Mex Helicase *Methylobacterium extorquens*
 AM1 Alphaproteobacteria Mex TrbC *Methylobacterium extorquens* AM1 Alphaproteobacteria Mfa
 RecA *Mycobacterium fallax* CITP8139, taxon: 1793 Mfl GyrA *Mycobacterium flavescens* Fla0
 taxon: 1776, reference #930991 Mfl RecA *Mycobacterium flavescens* Fla0 strain = Fla0, taxon:
 1776, ref. #930991 Mfl-ATCC14474 RecA *Mycobacterium flavescens*, strain = ATCC14474, taxon:
 1776, ATCC14474 ref #930991 Mfl-PYR-GCK DnaB *Mycobacterium flavescens* PYR- taxon:
 350054 GCK Mga GyrA *Mycobacterium gastri* HP4389, taxon: 1777 Mga RecA *Mycobacterium*
gastri HP4389, taxon: 1777 Mga SufB (Mga Pps1) *Mycobacterium gastri* HP4389, taxon: 1777
 Mgi-PYR-GCK DnaB *Mycobacterium gilvum* PYR-GCK taxon: 350054 Mgi-PYR-GCK GyrA
Mycobacterium gilvum PYR-GCK taxon: 350054 Mgo GyrA *Mycobacterium gordonae* taxon:
 1778, reference number 930835 Min-1442 DnaB *Mycobacterium intracellulare* strain 1442, taxon:
 1767 Min-ATCC13950 *Mycobacterium intracellulare* ATCC Taxon: 487521 GyrA 13950 Mkas
 GyrA *Mycobacterium kansasii* taxon: 1768 Mkas-ATCC12478 *Mycobacterium kansasii* ATCC
 Taxon: 557599 GyrA 12478 Mle-Br4923 GyrA *Mycobacterium leprae* Br4923 Taxon: 561304 Mle-
 TN DnaB *Mycobacterium leprae*, strain TN Human pathogen, taxon: 1769 Mle-TN GyrA
Mycobacterium leprae TN Human pathogen, STRAIN = TN, taxon: 1769 Mle-TN RecA
Mycobacterium leprae, strain TN Human pathogen, taxon: 1769 Mle-TN SufB (Mle
Mycobacterium leprae Human pathogen, taxon: 1769 Pps1) Mma GyrA *Mycobacterium*
malmoense taxon: 1780 Mmag Magn8951 BIL *Magnetospirillum magnetotacticum* Gram negative,
 taxon: 272627 MS-1 Msh RecA *Mycobacterium shimodei* ATCC27962, taxon: 29313 Msm DnaB-1
Mycobacterium smegmatis MC2 155 MC2 155, taxon: 246196 Msm DnaB-2 *Mycobacterium*

smegmatis MC2 155 MC2 155, taxon: 246196 Msp-KMS DnaB *Mycobacterium* species KMS
 taxon: 189918 Msp-KMS GyrA *Mycobacterium* species KMS taxon: 189918 Msp-MCS DnaB
Mycobacterium species MCS taxon: 164756 Msp-MCS GyrA *Mycobacterium* species MCS taxon:
 164756 Mthe RecA *Mycobacterium thermoresistibile* ATCC19527, taxon: 1797 Mtu SufB (Mtu
 Pps1) *Mycobacterium tuberculosis* strains Human pathogen, taxon: 83332 H37Rv & CDC1551
 Mtu-C RecA *Mycobacterium tuberculosis* C Taxon: 348776 Mtu-CDC1551 DnaB *Mycobacterium*
tuberculosis, Human pathogen, taxon: 83332 CDC1551 Mtu-CPHL RecA *Mycobacterium*
tuberculosis Taxon: 611303 CPHL_A Mtu-Canetti RecA *Mycobacterium tuberculosis*/ Taxon: 1773
 strain = “Canetti” Mtu-EAS054 RecA *Mycobacterium tuberculosis* EAS054 Taxon: 520140 Mtu-
 F11 DnaB *Mycobacterium tuberculosis*, strain taxon: 336982 F11 Mtu-H37Ra DnaB
Mycobacterium tuberculosis H37Ra ATCC 25177, taxon: 419947 Mtu-H37Rv DnaB
Mycobacterium tuberculosis H37Rv Human pathogen, taxon: 83332 Mtu-H37Rv RecA
Mycobacterium tuberculosis Human pathogen, taxon: 83332 H37Rv, Also CDC1551 Mtu-Haarlem
 DnaB *Mycobacterium tuberculosis* str. Taxon: 395095 Haarlem Mtu-K85 RecA *Mycobacterium*
tuberculosis K85 Taxon: 611304 Mtu-R604 RecA-n *Mycobacterium tuberculosis* ‘98- Taxon:
 555461 R604 INH-RIF-EM’ Mtu-So93 RecA *Mycobacterium tuberculosis* Human pathogen,
 taxon: 1773 So93/sub_species = “Canetti” Mtu-T17 RecA-c *Mycobacterium tuberculosis* T17
 Taxon: 537210 Mtu-T17 RecA-n *Mycobacterium tuberculosis* T17 Taxon: 537210 Mtu-T46 RecA
Mycobacterium tuberculosis T46 Taxon: 611302 Mtu-T85 RecA *Mycobacterium tuberculosis* T85
 Taxon: 520141 Mtu-T92 RecA *Mycobacterium tuberculosis* T92 Taxon: 515617 Mvan DnaB
Mycobacterium vanbaalenii PYR-1 taxon: 350058 Mvan GyrA *Mycobacterium vanbaalenii* PYR-1
 taxon: 350058 Mxa RAD25 *Myxococcus xanthus* DK1622 Deltaproteobacteria Mxe GyrA
Mycobacterium xenopi strain taxon: 1789 IMM5024 Naz-0708 RIR1-1 *Nostoc azollae* 0708 Taxon:
 551115 Naz-0708 RIR1-2 *Nostoc azollae* 0708 Taxon: 551115 Nfa DnaB *Nocardia farcinica* IFM
 10152 taxon: 247156 Nfa Nfa15250 *Nocardia farcinica* IFM 10152 taxon: 247156 Nfa RIR1
Nocardia farcinica IFM 10152 taxon: 247156 Nosp-CCY9414 DnaE-n *Nodularia spumigena*
 CCY9414 Taxon: 313624 Npu DnaB *Nostoc punctiforme* *Cyanobacterium*, taxon: 63737 Npu
 GyrB *Nostoc punctiforme* *Cyanobacterium*, taxon: 63737 Npu-PCC73102 DnaE-c *Nostoc*
punctiforme PCC73102 *Cyanobacterium*, taxon: 63737, ATCC29133 Npu-PCC73102 DnaE-n
Nostoc punctiforme PCC73102 *Cyanobacterium*, taxon: 63737, ATCC29133 Nsp-JS614 DnaB
Nocardioides species JS614 taxon: 196162 Nsp-JS614 TOPRIM *Nocardioides* species JS614
 taxon: 196162 Nsp-PCC7120 DnaB *Nostoc* species PCC7120, (*Anabaena* *Cyanobacterium*,
 Nitrogen- sp. PCC7120) fixing, taxon: 103690 Nsp-PCC7120 DnaE-c *Nostoc* species PCC7120,
 (*Anabaena* *Cyanobacterium*, Nitrogen- sp. PCC7120) fixing, taxon: 103690 Nsp-PCC7120 DnaE-n
Nostoc species PCC7120, (*Anabaena* *Cyanobacterium*, Nitrogen- sp. PCC7120) fixing, taxon:
 103690 Nsp-PCC7120 RIR1 *Nostoc* species PCC7120, (*Anabaena* *Cyanobacterium*, Nitrogen- sp.
 PCC7120) fixing, taxon: 103690 Oli DnaE-c *Oscillatoria limnetica* str. ‘Solar Lake’
Cyanobacterium, taxon: 262926 Oli DnaE-n *Oscillatoria limnetica* str. ‘Solar Lake’
Cyanobacterium, taxon: 262926 PP-PhiEL Helicase *Pseudomonas aeruginosa* phage Phage infects
Pseudomonas phiEL *aeruginosa*, taxon: 273133 PP-PhiEL ORF11 *Pseudomonas aeruginosa* phage
 phage infects *Pseudomonas* phiEL *aeruginosa*, taxon: 273133 PP-PhiEL ORF39 *Pseudomonas*
aeruginosa phage Phage infects *Pseudomonas* phiEL *aeruginosa*, taxon: 273133 PP-PhiEL ORF40
Pseudomonas aeruginosa phage phage infects *Pseudomonas* phiEL *aeruginosa*, taxon: 273133 Pfl
 Fha BIL *Pseudomonas fluorescens* Pf-5 Plant commensal organism, taxon: 220664 Plut RIR1
Pelodictyon luteolum DSM 273 Green sulfur bacteria, Taxon 319225 Pma-EXH1 GyrA
Persephonella marina EX-H1 Taxon: 123214 Pma-ExH1 DnaE *Persephonella marina* EX-H1
 Taxon: 123214 Pna RIR1 *Polaromonas naphthalenivorans* CJ2 taxon: 365044 Pnuc DnaB
Polynucleobacter sp. QLW- taxon: 312153 P1DMWA-1 Posp-JS666 DnaB *Polaromonas* species
 JS666 taxon: 296591 Posp-JS666 RIR1 *Polaromonas* species JS666 taxon: 296591 Pssp-A1-1 Fha
Pseudomonas species A1-1 Psy Fha *Pseudomonas syringae* pv. tomato Plant (tomato) pathogen,

str. DC3000 taxon: 223283 Rbr-D9 GyrB *Raphidiopsis brookii* D9 Taxon: 533247 Rce RIR1
Rhodospirillum centenum SW taxon: 414684, ATCC 51521 Rer-SK121 DnaB *Rhodococcus*
erythropolis SK121 Taxon: 596309 Rma DnaB *Rhodothermus marinus* Thermophile, taxon: 29549
 Rma-DSM4252 DnaB *Rhodothermus marinus* DSM 4252 Taxon: 518766 Rma-DSM4252 DnaE
Rhodothermus marinus DSM 4252 Thermophile, taxon: 518766 Rsp RIR1 *Roseovarius* species
 217 taxon: 314264 SaP-SETP12 dpol *Salmonella* phage SETP12 Phage, taxon: 424946 SaP-SETP3
 Helicase *Salmonella* phage SETP3 Phage, taxon: 424944 SaP-SETP3 dpol *Salmonella* phage
 SETP3 Phage, taxon: 424944 SaP-SETP5 dpol *Salmonella* phage SETP5 Phage, taxon: 424945
 Sare DnaB *Salinispora arenicola* CNS-205 taxon: 391037 Sav RecG Helicase *Streptomyces*
avermitilis MA-4680 taxon: 227882, ATCC 31267 Sel-PC6301 RIR1 *Synechococcus elongatus*
 PCC 6301 taxon: 269084 Berkely strain 6301~equivalent name: Ssp PCC 6301~synonym:
Anacystis nudulans Sel-PC7942 DnaE-c *Synechococcus elongatus* PC7942 taxon: 1140 Sel-
 PC7942 DnaE-n *Synechococcus elongatus* PC7942 taxon: 1140 Sel-PC7942 RIR1 *Synechococcus*
elongatus PC7942 taxon: 1140 Sel-PCC6301 DnaE-c *Synechococcus elongatus* PCC 6301
Cyanobacterium, and PCC7942 taxon: 269084, “Berkely strain 6301~equivalent name:
Synechococcus sp. PCC 6301~synonym: *Anacystis nudulans*” Sel-PCC6301 DnaE-n
Synechococcus elongatus PCC 6301 *Cyanobacterium*, taxon: 269084 “Berkely strain
 6301~equivalent name: *Synechococcus* sp. PCC 6301~synonym: *Anacystis nudulans*” Sep RIR1
Staphylococcus epidermidis RP62A taxon: 176279 ShP-Sfv-2a-2457T-n *Shigella flexneri* 2a str.
 2457T Putative bacteriophage Primase ShP-Sfv-2a-301-n *Shigella flexneri* 2a str. 301 Putative
 bacteriophage Primase ShP-Sfv-5 Primase *Shigella flexneri* 5 str. 8401 Bacteriophage,
 isolation_source_epidemic, taxon: 373384 SoP-SO1 dpol *Sodalis* phage SO-1
 Phage/isolation_source = “*Sodalis glossinidius* strain GA-SG, secondary symbiont of *Glossina*
austeni (Newstead)” Spl DnaX *Spirulina platensis*, strain C1 *Cyanobacterium*, taxon: 1156 Sru
 DnaB *Salinibacter ruber* DSM 13855 taxon: 309807, strain = “DSM 13855; M31” Sru PolBc
Salinibacter ruber DSM 13855 taxon: 309807, strain = “DSM 13855; M31” Sru RIR1 *Salinibacter*
ruber DSM 13855 taxon: 309807, strain = “DSM 13855; M31” Ssp DnaB *Synechocystis* species,
 strain *Cyanobacterium*, taxon: 1148 PCC6803 Ssp DnaE-c *Synechocystis* species, strain
Cyanobacterium, taxon: 1148 PCC6803 Ssp DnaE-n *Synechocystis* species, strain *Cyanobacterium*,
 taxon: 1148 PCC6803 Ssp DnaX *Synechocystis* species, strain *Cyanobacterium*, taxon: 1148
 PCC6803 Ssp GyrB *Synechocystis* species, strain *Cyanobacterium*, taxon: 1148 PCC6803 Ssp-JA2
 DnaB *Synechococcus* species JA-2-3B'a(2-13) *Cyanobacterium*, Taxon: 321332 Ssp-JA2 RIR1
Synechococcus species JA-2-3B'a(2-13) *Cyanobacterium*, Taxon: 321332 Ssp-JA3 DnaB
Synechococcus species JA-3-3Ab *Cyanobacterium*, Taxon: 321327 Ssp-JA3 RIR1 *Synechococcus*
 species JA-3-3Ab *Cyanobacterium*, Taxon: 321327 Ssp-PCC7002 DnaE-c *Synechocystis* species,
 strain PCC *Cyanobacterium*, taxon: 32049 7002 Ssp-PCC7002 DnaE-n *Synechocystis* species,
 strain PCC *Cyanobacterium*, taxon: 32049 7002 Ssp-PCC7335 RIR1 *Synechococcus* sp. PCC 7335
 Taxon: 91464 StP-Twort ORF6 *Staphylococcus* phage Twort Phage, taxon 55510 Susp-NBC371
 DnaB *Sulfurovum* sp. NBC37-1 taxon: 387093 intein Taq-Y51MC23 DnaE *Thermus aquaticus*
 Y51MC23 Taxon: 498848 Taq-Y51MC23 RIR1 *Thermus aquaticus* Y51MC23 Taxon: 498848
 Tcu-DSM43183 RecA *Thermomonospora curvata* DSM Taxon: 471852 43183 Tel DnaE-c
Thermosynechococcus elongatus BP-1 *Cyanobacterium*, taxon: 197221 Tel DnaE-n
Thermosynechococcus elongatus BP-1 *Cyanobacterium*, Ter DnaB-1 *Trichodesmium erythraeum*
 IMS101 *Cyanobacterium*, taxon: 203124 Ter DnaB-2 *Trichodesmium erythraeum* IMS101
Cyanobacterium, taxon: 203124 Ter DnaE-1 *Trichodesmium erythraeum* IMS101 *Cyanobacterium*,
 taxon: 203124 Ter DnaE-2 *Trichodesmium erythraeum* IMS101 *Cyanobacterium*, taxon: 203124
 Ter DnaE-3c *Trichodesmium erythraeum* IMS101 *Cyanobacterium*, taxon: 203124 Ter DnaE-3n
Trichodesmium erythraeum IMS101 *Cyanobacterium*, taxon: 203124 Ter GyrB *Trichodesmium*
erythraeum IMS101 *Cyanobacterium*, taxon: 203124 Ter Ndse-1 *Trichodesmium erythraeum*
 IMS101 *Cyanobacterium*, taxon: 203124 Ter Ndse-2 *Trichodesmium erythraeum* IMS101

Cyanobacterium, taxon: 203124 Ter RIR1-1 *Trichodesmium erythraeum* IMS101 Cyanobacterium, taxon: 203124 Ter RIR1-2 *Trichodesmium erythraeum* IMS101 Cyanobacterium, taxon: 203124 Ter RIR1-3 *Trichodesmium erythraeum* IMS101 Cyanobacterium, taxon: 203124 Ter RIR1-4 *Trichodesmium erythraeum* IMS101 Cyanobacterium, taxon: 203124 Ter Snf2 *Trichodesmium erythraeum* IMS101 Cyanobacterium, taxon: 203124 Ter ThyX *Trichodesmium erythraeum* IMS101 Cyanobacterium, taxon: 203124 Tfus RecA-1 *Thermobifida fusca* YX Thermophile, taxon: 269800 Tfus RecA-2 *Thermobifida fusca* YX Thermophile, taxon: 269800 Tfus Tfu2914 *Thermobifida fusca* YX Thermophile, taxon: 269800 Thsp-K90 RIR1 *Thioalkalivibrio* sp. K90mix Taxon: 396595 Tth-DSM571 RIR1 *Thermoanaerobacterium* Taxon: 580327 *thermosaccharolyticum* DSM 571 Tth-HB27 DnaE-1 *Thermus thermophilus* HB27 thermophile, taxon: 262724 Tth-HB27 DnaE-2 *Thermus thermophilus* HB27 thermophile, taxon: 262724 Tth-HB27 RIR1-1 *Thermus thermophilus* HB27 thermophile, taxon: 262724 Tth-HB27 RIR1-2 *Thermus thermophilus* HB27 thermophile, taxon: 262724 Tth-HB8 DnaE-1 *Thermus thermophilus* HB8 thermophile, taxon: 300852 Tth-HB8 DnaE-2 *Thermus thermophilus* HB8 thermophile, taxon: 300852 Tth-HB8 RIR1-1 *Thermus thermophilus* HB8 thermophile, taxon: 300852 Tth-HB8 RIR1-2 *Thermus thermophilus* HB8 thermophile, taxon: 300852 Tvu DnaE-c *Thermosynechococcus vulcanus* Cyanobacterium, taxon: 32053 Tvu DnaE-n *Thermosynechococcus vulcanus* Cyanobacterium, taxon: 32053 Tye RNR-1 *Thermodesulfovibrio yellowstonii* taxon: 289376 DSM 11347 Tye RNR-2 *Thermodesulfovibrio yellowstonii* taxon: 289376 DSM 11347 Archaea Ape APE0745 *Aeropyrum pernix* K1 Thermophile, taxon: 56636 Cme-boo Pol-II *Candidatus Methanoregula boonei* taxon: 456442 6A8 Fac-Fer1 RIR1 *Ferroplasma acidarmanus*, strain Fer1, eats iron taxon: 97393 and taxon 261390 Fac-Fer1 SufB (Fac *Ferroplasma acidarmanus* strain fer1, eats Pps1) iron, taxon: 97393 Fac-TypeI RIR1 *Ferroplasma acidarmanus* type I, Eats iron, taxon 261390 Fac-typeI SufB (Fac *Ferroplasma acidarmanus* Eats iron, taxon: 261390 Pps1) Hma CDC21 *Haloarcula marismortui* ATCC taxon: 272569, 43049 Hma Pol-II *Haloarcula marismortui* ATCC taxon: 272569, 43049 Hma PolB *Haloarcula marismortui* ATCC taxon: 272569, 43049 Hma TopA *Haloarcula marismortui* ATCC taxon: 272569 43049 Hmu-DSM12286 *Halomicrobium mukohataei* DSM taxon: 485914 (*Halobacteria*) MCM 12286 Hmu-DSM12286 PolB *Halomicrobium mukohataei* DSM Taxon: 485914 12286 Hsa-R1 MCM *Halobacterium salinarum* R-1 Halophile, taxon: 478009, strain = "R1; DSM 671" Hsp-NRC1 CDC21 *Halobacterium* species NRC-1 Halophile, taxon: 64091 Hsp-NRC1 Pol-II *Halobacterium salinarum* NRC-1 Halophile, taxon: 64091 Hut MCM-2 *Halorhabdus utahensis* DSM 12940 taxon: 519442 Hut-DSM12940 MCM-1 *Halorhabdus utahensis* DSM 12940 taxon: 519442 Hvo PolB *Haloferax volcanii* DS70 taxon: 2246 Hwa GyrB *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa MCM-1 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa MCM-2 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa MCM-3 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa MCM-4 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa Pol-II-1 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa Pol-II-2 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa PolB-1 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa PolB-2 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa PolB-3 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa RCF *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa RIR1-1 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa RIR1-2 *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa Top6B *Haloquadratum walsbyi* DSM 16790 Halophile, taxon: 362976, strain: DSM 16790 = HBSQ001 Hwa rPol A" *Haloquadratum walsbyi* DSM 16790 Halophile, taxon:

362976, strain: DSM 16790 = HBSQ001 Maeo Pol-II *Methanococcus aeolicus* Nankai-3 taxon: 419665 Maeo RFC *Methanococcus aeolicus* Nankai-3 taxon: 419665 Maeo RNR *Methanococcus aeolicus* Nankai-3 taxon: 419665 Maeo-N3 Helicase *Methanococcus aeolicus* Nankai-3 taxon: 419665 Maeo-N3 RtcB *Methanococcus aeolicus* Nankai-3 taxon: 419665 Maeo-N3 UDP GD *Methanococcus aeolicus* Nankai-3 taxon: 419665 Mein-ME PEP *Methanocaldococcus infernus* ME thermophile, Taxon: 573063 Mein-ME RFC *Methanocaldococcus infernus* ME Taxon: 573063 Memar MCM2 *Methanoculleus marisnigri* JR1 taxon: 368407 Memar Pol-II *Methanoculleus marisnigri* JR1 taxon: 368407 Mesp-FS406 PolB-1 *Methanocaldococcus* sp. FS406-22 Taxon: 644281 Mesp-FS406 PolB-2 *Methanocaldococcus* sp. FS406-22 Taxon: 644281 Mesp-FS406 PolB-3 *Methanocaldococcus* sp. FS406-22 Taxon: 644281 Mesp-FS406-22 LHR *Methanocaldococcus* sp. FS406-22 Taxon: 644281 Mfe-AG86 Pol-1 *Methanocaldococcus fervens* AG86 Taxon: 573064 Mfe-AG86 Pol-2 *Methanocaldococcus fervens* AG86 Taxon: 573064 Mhu Pol-II *Methanospirillum hungatei* JF-1 taxon 323259 Mja GF-6P *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja Helicase *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja Hyp-1 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja IF2 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja KlbA *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja PEP *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja Pol-1 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja Pol-2 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja RFC-1 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja RFC-2 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja RFC-3 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja RNR-1 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja RNR-2 *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja RtcB (Mja Hyp-2) *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja TFIIB *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja UDP GD *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja r-Gyr *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja rPol A' *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mja rPol A'' *Methanococcus jannaschii* Thermophile, DSM 2661, (*Methanocaldococcus jannaschii* taxon: 2190 DSM 2661) Mka CDC48 *Methanopyrus kandleri* AV19 Thermophile, taxon: 190192 Mka EF2 *Methanopyrus kandleri* AV19 Thermophile, taxon: 190192 Mka RFC *Methanopyrus kandleri* AV19 Thermophile, taxon: 190192 Mka RtcB *Methanopyrus kandleri* AV19 Thermophile, taxon: 190192 Mka VatB *Methanopyrus kandleri* AV19 Thermophile, taxon: 190192 Mth RIR1 *Methanothermobacter* Thermophile, delta H strain *thermautotrophicus* (*Methanobacterium thermoautotrophicum*) Mvu-M7 Helicase *Methanocaldococcus vulcanius* M7 Taxon: 579137 Mvu-M7 Pol-1 *Methanocaldococcus vulcanius* M7 Taxon: 579137 Mvu-M7 Pol-2 *Methanocaldococcus vulcanius* M7 Taxon: 579137 Mvu-M7 Pol-3 *Methanocaldococcus vulcanius* M7 Taxon: 579137 Mvu-M7 UDP GD *Methanocaldococcus vulcanius* M7 Taxon: 579137 Neq Pol-c *Nanoarchaeum equitans* Kin4-M Thermophile, taxon: 228908 Neq Pol-n *Nanoarchaeum equitans* Kin4-M Thermophile, taxon: 228908 Nma-ATCC43099 *Natrialba magadii* ATCC 43099 Taxon: 547559 MCM Nma-ATCC43099 *Natrialba magadii* ATCC 43099 Taxon: 547559 PolB-1 Nma-ATCC43099 *Natrialba magadii* ATCC 43099

Taxon: 547559 PolB-2 Nph CDC21 *Natronomonas pharaonis* DSM 2160 taxon: 348780 Nph PolB-1 *Natronomonas pharaonis* DSM 2160 taxon: 348780 Nph PolB-2 *Natronomonas pharaonis* DSM 2160 taxon: 348780 Nph rPol A" *Natronomonas pharaonis* DSM 2160 taxon: 348780 Pab CDC21-1 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab CDC21-2 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab IF2 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab KlbA *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab Lon *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab Moaa *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab Pol-II *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab RFC-1 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab RFC-2 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab RIR1-1 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab RIR1-2 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab RIR1-3 *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab RtcB (Pab Hyp-2) *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Pab VMA *Pyrococcus abyssi* Thermophile, strain Orsay, taxon: 29292 Par RIR1 *Pyrobaculum arsenaticum* DSM taxon: 340102 13514 Pfu CDC21 *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu IF2 *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu KlbA *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu Lon *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu RFC *Pyrococcus furiosus* Thermophile, DSM3638, taxon: 186497 Pfu RIR1-1 *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu RIR1-2 *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu RtcB (Pfu Hyp-2) *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu TopA *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pfu VMA *Pyrococcus furiosus* Thermophile, taxon: 186497, DSM3638 Pho CDC21-1 *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho CDC21-2 *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho IF2 *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho KlbA *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho LHR *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho Lon *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho Pol I *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho Pol-II *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho RFC *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho RIR1 *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho RadA *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho RtcB (Pho Hyp-2) *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho VMA *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Pho r-Gyr *Pyrococcus horikoshii* OT3 Thermophile, taxon: 53953 Psp-GBD Pol *Pyrococcus* species GB-D Thermophile Pto VMA *Picrophilus torridus*, DSM 9790 DSM 9790, taxon: 263820, Thermoacidophile Smar 1471 *Staphylothermus marinus* F1 taxon: 399550 Smar MCM2 *Staphylothermus marinus* F1 taxon: 399550 Tac-ATCC25905 VMA *Thermoplasma acidophilum*, ATCC Thermophile, taxon: 2303 25905 Tac-DSM1728 VMA *Thermoplasma acidophilum*, Thermophile, taxon: 2303 DSM1728 Tag Pol-1 (Tsp-TY Pol-1) *Thermococcus aggregans* Thermophile, taxon: 110163 Tag Pol-2 (Tsp-TY Pol-2) *Thermococcus aggregans* Thermophile, taxon: 110163 Tag Pol-3 (Tsp-TY Pol-3) *Thermococcus aggregans* Thermophile, taxon: 110163 Tba Pol-II *Thermococcus barophilus* MP taxon: 391623 Tfu Pol-1 *Thermococcus fumicolans* Thermophile, taxon: 46540 Tfu Pol-2 *Thermococcus fumicolans* Thermophile, taxon: 46540 Thy Pol-1 *Thermococcus hydrothermalis* Thermophile, taxon: 46539 Thy Pol-2 *Thermococcus hydrothermalis* Thermophile, taxon: 46539 Tko CDC21-1 *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko CDC21-2 *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko Helicase *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko IF2 *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko KlbA *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko LHR *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko Pol-1 (Pko Pol-1) *Pyrococcus/Thermococcus* Thermophile, taxon: 69014 *kodakaraensis* KOD1 Tko Pol-2 (Pko Pol-2) *Pyrococcus/Thermococcus* Thermophile, taxon: 69014 *kodakaraensis* KOD1 Tko Pol-II *Thermococcus kodakaraensis* KOD1 Thermophile,

taxon: 69014 Tko RFC *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko RIR1-1 *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko RIR1-2 *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko RadA *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko TopA *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tko r-Gyr *Thermococcus kodakaraensis* KOD1 Thermophile, taxon: 69014 Tli Pol-1 *Thermococcus litoralis* Thermophile, taxon: 2265 Tli Pol-2 *Thermococcus litoralis* Thermophile, taxon: 2265 Tma Pol *Thermococcus marinus* taxon: 187879 Ton-NA1 LHR *Thermococcus onnurineus* NA1 Taxon: 523850 Ton-NA1 Pol *Thermococcus onnurineus* NA1 taxon: 342948 Tpe Pol *Thermococcus peptonophilus* strain taxon: 32644 SM2 Tsi-MM739 Lon *Thermococcus sibiricus* MM 739 Thermophile, Taxon: 604354 Tsi-MM739 Pol-1 *Thermococcus sibiricus* MM 739 Taxon: 604354 Tsi-MM739 Pol-2 *Thermococcus sibiricus* MM 739 Taxon: 604354 Tsi-MM739 RFC *Thermococcus sibiricus* MM 739 Taxon: 604354 Tsp AM4 RtcB *Thermococcus* sp. AM4 Taxon: 246969 Tsp-AM4 LHR *Thermococcus* sp. AM4 Taxon: 246969 Tsp-AM4 Lon *Thermococcus* sp. AM4 Taxon: 246969 Tsp-AM4 RIR1 *Thermococcus* sp. AM4 Taxon: 246969 Tsp-GE8 Pol-1 *Thermococcus* species GE8 Thermophile, taxon: 105583 Tsp-GE8 Pol-2 *Thermococcus* species GE8 Thermophile, taxon: 105583 Tsp-GT Pol-1 *Thermococcus* species GT taxon: 370106 Tsp-GT Pol-2 *Thermococcus* species GT taxon: 370106 Tsp-OGL-20P Pol *Thermococcus* sp. OGL-20P taxon: 277988 Tthi Pol *Thermococcus thio-reducens* Hyperthermophile Tvo VMA *Thermoplasma volcanium* GSS1 Thermophile, taxon: 50339 Tzi Pol *Thermococcus zilligii* taxon: 54076 Unc-ERS PFL uncultured archaeon GZfos13E1 isolation_source = "Eel River sediment", clone = "GZfos13E1", taxon: 285397 Unc-ERS RIR1 uncultured archaeon GZfos9C4 isolation_source = "Eel River sediment", taxon: 285366, clone = "GZfos9C4" Unc-ERS RNR uncultured archaeon GZfos10C7 isolation_source = "Eel River sediment", clone = "GZfos10C7", taxon: 285400 Unc-MetRFS MCM2 uncultured archaeon (Rice Cluster I) Enriched methanogenic consortium from rice field soil, taxon: 198240

[0069] The split inteins of the disclosed compositions or that can be used in the disclosed methods can be modified, or mutated, inteins. A modified intein can comprise modifications to the N-terminal intein segment, the C-terminal intein segment, or both. The modifications can include additional amino acids at the N-terminus the C-terminus of either portion of the split intein, or can be within the either portion of the split intein. Table 2 shows a list of amino acids, their abbreviations, polarity, and charge.

TABLE-US-00002

TABLE 2	List of Amino Acids	3-Letter	1-Letter	Amino Acid	Code	Code
Polarity	Charge	Alanine	Ala	A	nonpolar	neutral
Arginine	Arg	R	Basic	positive	polar	Asparagine
Asn	N	polar	neutral	Aspartic acid	Asp	D
acidic	negative	polar	Cysteine	Cys	C	nonpolar
neutral	Glutamic acid	Glu	E	acidic	negative	polar
Glutamine	Gln	Q	polar	neutral	Glycine	Gly
G	nonpolar	Histidine	His	H	Basic	Positive (10%)
polar	Neutral (90%)	Isoleucine	Ile	I	nonpolar	neutral
Leucine	Leu	L	nonpolar	neutral	Lysine	Lys
K	Basic	positive	polar	Methionine	Met	M
nonpolar	neutral	Phenylalanine	Phe	F	nonpolar	neutral
Proline	Pro	P	nonpolar	neutral	Serine	Ser
S	polar	Threonine	Thr	T	polar	neutral
Tryptophan	Trp	W	nonpolar	neutral	Tyrosine	Tyr
Y	polar	Valine	Val	V	nonpolar	neutral

[0070] The N-intein of the invention may be coupled to solid phase, such as a membrane, fiber, particle, bead or chip. The solid phase may be a chromatography resin of natural or synthetic origin, such as a natural or synthetic resin, preferably a polysaccharide such as agarose. The solid phase, such as a chromatography resin, may be provided with embedded magnetic particles. In another embodiment the solid phase is a non-diffusion limited resin/fibrous material.

[0071] In this case the solid phase may be formed from one or more polymeric nanofibre substrates, such as electrospun polymer nanofibres. Polymer nanofibres for use in the present invention typically have mean diameters from 10 nm to 1000 nm. The length of polymer nanofibres is not particularly limited. The polymer nanofibres can suitably be monofilament nanofibres and may e.g. have a circular, ellipsoidal or essentially circular/ellipsoidal cross section. Typically, the

one or more polymer nanofibres are provided in the form of one or more non-woven sheets, each comprising one or more polymer nanofibers. A non-woven sheet comprising one or more polymer nanofibres is a mat of said one or more polymer nanofibres with each nanofibre oriented essentially randomly, i.e. it has not been fabricated so that the nanofibre or nanofibres adopts a particular pattern. Non-woven sheets typically have area densities from 1 to 40 g/m². Non-woven sheets typically have a thickness from 5 to 120 µm. The polymer should be a polymer suitable for use as a chromatography medium, i.e. an adsorbent, in a chromatography method. Suitable polymers include polyamides such as nylon, polyacrylic acid, polymethacrylic acid, polyacrylonitrile, polystyrene, polysulfones e.g. polyethersulfone (PES), polycaprolactone, collagen, chitosan, polyethylene oxide, agarose, agarose acetate, cellulose, cellulose acetate, and combinations thereof. [0072] The N-intein according to the invention may be immobilized on a solid support in a very high degree, 0.2-2 µmole/ml N-intein is coupled per ml resin (swollen gel).

[0073] The N-intein according to the invention may be coupled to the solid phase via a Lys-tail, comprising one or more Lys, such as at least two, on the C-terminal. Alternatively, the N-intein is coupled to the solid phase via a Cys-tail on the C-terminal.

C-Intein Protein Variants

[0074] Preferably the invention also provides a C-intein comprising a split intein C-intein sequence or engineered variants thereof.

[0075] It will be appreciated that selection of the N-intein and C-intein can be from the same wild type split intein (e.g., both from Npu, or a variant of either the N- or C-intein, or alternatively can be selected from different wild type split inteins or the consensus split intein sequences, as it has been discovered that the affinity of a N-fragment for a different C-fragment (e.g., Npu N-fragment or variant thereof with Ssp C-fragment or variant thereof) still maintains sufficient binding affinity for use in the disclosed methods.

Split Intein Systems

[0076] Preferably, the invention provides a split intein system for affinity purification of a protein of interest (POI), comprising a N-intein and C-intein as described above.

[0077] Preferably the N-intein is attached to a solid phase and the C-intein is co-expressed with the POI and used as a tag for affinity purification of the POI. Vice versa is also possible, ie attaching the C-intein to a solid phase and using the N-intein as a tag, but the former is preferred.

[0078] In one embodiment the C-intein and an additional tag is co-expressed with the POI. The additional tag may be any conventional chromatography tag, such as an IEX tag or an affinity tag.

Methods of Purifying a Protein of Interest (POI)

[0079] The invention relates to a method for purification of a protein of interest (POI), using the split intein system according to the invention, comprising association of the C-intein and N-intein at neutral pH, such as 6-8, and in the presence of divalent cations (which impairs spontaneous cleavage); washing said solid phase in the presence of divalent cations; addition of a chelator to allow spontaneous cleavage between C-intein and POI; collection of tagless POI.

[0080] This protocol is suitable for protein non-sensitive for Zn. The advantages are long contact times are allowed with the resin and addition of large sample volume. Sample loading could be made for long times, such as up to 1.5 hours.

[0081] According to the invention more than 30% yield, preferably 50%, most preferably more than 80% of POI is achieved in less than 4 hours cleavage.

[0082] The invention enables a high ligand density when the N-intein is immobilized to a solid phase. Preferably the N-intein is attached to a chromatography resin, such as agarose or any other suitable resin for protein purification. According to the invention it is possible to achieve a static binding capacity of 0.2-2 µmole/ml C-intein bound POI per settled ml resin.

Affinity Tags

[0083] The invention also relates to a method for purification of a protein of interest (POI), comprising the following steps: co-expressing a POI with a C-intein according to the invention and

an additional tag; binding said additional tag to its binding partner on a solid phase; cleaving off the POI and the C-intein; binding said C-intein to an N-intein attached to a solid phase at neutral pH and cleaving off said bound C-intein and N-intein from said POI; and re-generating said solid phase under alkaline conditions, such as 0.5M NaOH. The purpose of this twin tag: increased purity (enables dual affinity purification), solubility, detectability.

[0084] Affinity tags can be peptide or protein sequences cloned in frame with protein coding sequences that change the protein's behavior. Affinity tags can be appended to the N- or C-terminus of proteins which can be used in methods of purifying a protein from cells. Cells expressing a peptide comprising an affinity tag can be expressed with a signal sequence in the supernatant/cell culture medium. Cells expressing a peptide comprising an affinity tag can also be pelleted, lysed, and the cell lysate applied to a column, resin or other solid support that displays a ligand to the affinity tags. The affinity tag and any fused peptides are bound to the solid support, which can also be washed several times with buffer to eliminate unbound (contaminant) proteins. A protein of interest, if attached to an affinity tag, can be eluted from the solid support via a buffer that causes the affinity tag to dissociate from the ligand resulting in a purified protein, or can be cleaved from the bound affinity tag using a soluble protease. As disclosed herein, the affinity tag is cleaved through the self-cleaving mechanism of the C-intein segment in the active intein complex.

[0085] Examples of affinity include, but are not limited to, maltose binding protein, which can bind to immobilized maltose to facilitate purification of the fused target protein; Chitin binding protein, which can bind to immobilized chitin; Glutathione S transferase, which can bind to immobilized glutathione; poly-histidine, which can bind to immobilized chelated metals; FLAG octapeptide, which can bind to immobilized anti-FLAG antibodies.

[0086] Affinity tags can also be used to facilitate the purification of a protein of interest using the disclosed modified peptides through a variety of methods, including, but not limited to, selective precipitation, ion exchange chromatography, binding to precipitation-capable ligands, dialysis (by changing the size and/or charge of the target protein) and other highly selective separation methods.

[0087] In some aspects, affinity tags can be used that do not actually bind to a ligand, but instead either selectively precipitate or act as ligands for immobilized corresponding binding domains. In these instances, the tags are more generally referred to as purification tags. For example, the ELP tag selectively precipitates under specific salt and temperature conditions, allowing fused peptides to be purified by centrifugation. Another example is the antibody Fc domain, which serves as a ligand for immobilized protein A or Protein G-binding domains.

Proteins of Interest

[0088] Target proteins for all protocols are: any recombinant proteins, especially proteins requiring native or near native N-terminal sequences, for example therapeutic protein candidates, biologics, antibody fragments, antibody mimetics, protein scaffolds, enzymes, recombinant proteins or peptides, such as growth factors, cytokines, chemokines, hormones, antigen (viral, bacterial, yeast, mammalian) production, vaccine production, cell surface receptors, fusion proteins.

[0089] The invention will now be described more closely in association with some non-limiting examples and the accompanying drawings.

Experimental Part

[0090] The invention will be described more closely in association with some non-limiting examples and the accompanying drawings.

[0091] In the present invention the following 5 constructs were evaluated:

TABLE-US-00003 TABLE 1 A52

ALSYETEILTVEYGLLPKIVEKRIECTVYSVDNNGNIYTQPVAQWHDRGEQEVFEYCL
A53

ALSYDTEILTVEYGFLPKIVEENIECTVYSVDKNGFVYTQPIAQWHNRGEQEVFEYDL
B97_K24E/R25N

ALSYETEILTVEYGLLPKIVEENIECTVYSVDNNGNIYTQPVAQWHDRGEQEVFEYCL

B82_K24E
 ALSYETEILTVEYGLLPiGKIVEERIECTVYSVDNNGNIYTQPVAQWHDGRGEQEVFEYCL
 B83_R25N
 ALSYETEILTVEYGLLPiGKIVEKNIECTVYSVDNNGNIYTQPVAQWHDGRGEQEVFEYCL
 ****.*****.*****.*****.*** :****.*****.***** * A52
 EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
 A53
 EDGSIIRATKDHKFMTTDGEMLPIDEIFEQGLDLKQVGGSGDYKDDDDKGGSGHHHHHH
 B97_K24E/R25N
 EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
 B82_K24E
 EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
 B83_R25N
 EDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRVGGSGDYKDDDDKGGSGHHHHHH
 ****.*****.***.*****.*** :*****.*****.*****

Non-Intein sequences

Construct Cultivation and Expression:

[0092] Start cultures were diluted 1:100 in 100 mL LB+neo in shake flasks (done in triplicates for all 5 constructs) and incubated at 37 until OD600 was 0.6-1. Once target OD was reached, the flasks were transferred to a cooled incubator (22 degrees) and 0.5 mM IPTG was added for induction over night (exactly 18 h). Induced cultivations were pelleted sequentially at 5000 g (+7 degrees) in 50 mL falcon tubes and pellets were weighed.

Solubility Testing

[0093] Weighed pellets (ca 0.8 grams) were resuspended in 20 volumes of 1× PBS. For each of the constructs; [0094] 1. 20 µL was saved for SDS-PAGE (Whole cell Lysate/WCL) [0095] 2. 5 mL was added to 50 mL falcon tube and [0096] a. centrifuged for 20 min at 5000 g (6 degrees) [0097] b. pellet was resuspended in 5 mL 8M Urea [0098] c. incubated by end-to-end for ~1 h at room temp [0099] d. Centrifuged at 20000 g for 20 min at 6 degrees [0100] e. 20 µL of supernatant was saved for SDS-PAGE (Urea) [0101] f. Rest of supernatant was used for Biacore CFCA analysis [0102] 3. 10 mL was added to to a 50 mL falcon tube and [0103] a. Sonicated on ice for 1 min on time (2 sec on/4 sec off pulses) at 30% amplitude [0104] b. Centrifuged at 20000 g for 20 min at 6 degrees [0105] c. 20 µL of supernatant was saved for SDS-PAGE (Sonic.) [0106] d. Rest of supernatant was used for Biacore CFCA analysis [0107] 4. 500 µL was added to a 1.5 mL tube and [0108] a. Centrifuged for 20 min at 5000 g at 6 degrees [0109] b. Resuspended pellet in 500 µL 1% NP40 buffer [0110] c. Incubated for 40 min with 1200 rpm skaking (Mathilda block) at room temperature [0111] d. Centrifuged for 20 min at 12000 g at 6 degrees [0112] e. Saved ca 250 µL of the supernatant [0113] f. 20 µL of supernatant was saved for SDS-PAGE (NP40) [0114] g. Rest of supernatant was used for BiaCore CFCA analysis [0115] To the 20 µL samples 40 µL SDS-PAGE buffer was added and boiled at 95 degrees for 5 min prior to loading the samples to a 15% gel.

Solubility was Evaluated by SDS-PAGE Analysis Samples and Calculated Accordingly:

[0116]

Extraction method densitometric signal/Whole cell lysate (WCL) densitometric signal*100%
 [0117] FIG. 1 shows SDS-PAGE analysis of representative supernatants after using different extraction techniques. 20 µL of supernatants were mixed with 40 µL of 2× Laemmli sample buffer and boiled for 5 minutes at 95 degrees celsius prior to loading on a 15% homogenous SDS-PAGE gel. Gel was electrophoresed for 1 h and 50 min at 600V and stained by coomassie for approximately two hours. After extensive destaining, gel was imaged using an Amersham AI600 imager.

[0118] FIG. 2 Shows solubility determined after densitometric evaluation of SDS-PAGE analysis. Extracts from three different cell-cultures for each construct were analysed and ligand band densitometry was measured using ImageQuant TL software. Solubility was calculated based on the following formula:

$$\text{Densitometry}(\text{extraction method } X) / \text{Densitometry}(WCL) * 100\% = \% \text{ solubility}$$

[0119] Bars show the average solubility of different extraction methods compared to whole cell lysate and the error bars show the standard deviation. All constructs showed significantly better solubility than A52 for the sonicated samples analysed. A53, B97 and B82 showed significantly higher solubility as compared to A52 samples. Statistical significance was determined by one-way ANOVA with Dunnet's post test using A52 as control sample. *p-value <0.05, ** p-value <0.01, *** p-value <0.001, **** p-value <0.0001.

TABLE-US-00004 Construct name NP40, mean solubility Sonication, mean solubility A52 7.4% 22.1% A53 67.6% 104.7% B97 37.3% 92.8% B82 44.3% 62.8% B83 12.4% 36.6%

[0120] Urea extraction method showed no significant difference as compared to WCL and independent of the construct analysed.

Biacore CFCA Analysis

[0121] Determination of the soluble N-intein ratio of the various protein extracts was further analyzed by SPR binding analysis using a FLAG-epitope (DYKDDDDK) as a detection-tag at the C-terminus of the constructs. Calibration-free concentration analysis, (CFCA), was done in a Biacore T200 instrument using a mouse monoclonal ANTI-FLAG M2 antibody. Sensor chips, CM5 series S were immobilized with the anti-FLAG antibody using an amine coupling kit. 10 mM sodium acetate pH 4.0 was used as immobilization buffer, HBS-EP+pH 7.4 as a running buffer and Glycine-HCl pH 2.5 as regeneration buffer. The immobilization levels were about 8000-10000 RU. Supernatant samples from the different extractions described above, were diluted from 150-5000 times in HBS-EP+running buffer before analysis by the CFCA method. The molecular weights of the different protein constructs ranged from 13.5-13.6 kDa and this was used to calculate the diffusion coefficient at 20° C., (1.13389E.sup.-10 m.sup.2/s). The default Biacore method for CFCA was used as a starting point for setting up the final method. Sample concentrations were determined by using the Biacore T200 evaluation software.

[0122] FIG. 3 shows N-intein concentrations in supernatants from different extracts determined by Biacore CFCA analysis. Extracts from three different cell-cultures for each construct were analysed. Bars show the average concentration and the error bars show the standard deviation.

[0123] N-intein concentration in the supernatants after extraction and clarification using different extraction methods is used for the calculation of soluble N-intein ratios. The NP40 detergent buffer causes a mild release of soluble proteins from the cells. Ultra-sonication is a mechanical extraction technique causing vigorous cell disruption, releasing soluble proteins. Urea at high concentration is a denaturing extraction method causing the release of both soluble proteins and insoluble proteins found in inclusion bodies from the cells. Boiling of the cell pellets in a SDS sample buffer causes complete solubilization of both soluble and insoluble N-intein and is used as the reference for total amount of expressed N-intein. A high N-intein concentration in supernatants after extraction with non-denaturing extraction methods, (NP40 and sonication) compared with denaturing methods, (Urea and SDS) indicate a high solubility. The CFCA analysis show that the A53 and B97 constructs have a high solubility whereas A52 has a very poor solubility, FIG. 4.

[0124] Statistical analysis show that the modified constructs B82, B83 and B97 are significantly more soluble compared with the non-modified A52 construct when using mild non-denaturing extraction methods.

Solubility Evaluated by SPR Binding Analysis is Calculated Accordingly:

[0125]

$$\text{Extraction method } N\text{-intein concentration} / \text{SDS extracted } N\text{-intein concentration} * 100\%$$

Results

[0126] Sodium dodecyl sulfate, SDS, is an ionic detergent that binds to proteins through ionic and hydrophobic interactions and solubilizes proteins by altering their secondary and tertiary structure. SDS is routinely used in polyacrylamide gel electrophoresis, (SDS-PAGE) to separate, characterize and quantify proteins. SDS has been used in these example experiments as a universal protein solubilizing reagent used for quantification of the total amount of protein in different extracts, both soluble and insoluble for subsequent separation, detection and quantification by densitometric analysis of SDS-PAGE gels and Biacore calibration free concentration analysis, CFCA. The concentration of different constructs in SDS solubilized sample extracts is normalized to 100% for comparison with the concentration of the different protein constructs derived in the supernatants after centrifugal clarification of extracts using different methods.

[0127] A mild method for extracting soluble proteins only is the use of a non-ionic detergent NP40. NP40 at 1% (w/v) is added to a Tris-HCl buffer, pH 7.5 containing 150 mM sodium chloride and is simply used by resuspending harvested bacterial cell pellets followed by mixing during 1 hour. After incubation the cell suspension is clarified by centrifugation to remove insoluble material.

[0128] Ultra-sonication or sonication, is an extraction method for proteins that uses mechanical energy from a probe to disintegrate cells for the release of soluble cell components. Cells are resuspended in a non-denaturing buffer like phosphate buffered saline, PBS at pH 7.4 to control the pH during the release of cellular components. Sonication is a very efficient and reliable tool for cell disintegration that allows for a complete control over the sonication parameters. This ensures a high selectivity on materials release and product purity. After sonication, the lysate is clarified by supernatant and the insoluble pellet is removed.

[0129] Chaotropic salts like Urea can be used for the release of both soluble and insoluble proteins from cells. Urea is compatible with a wide range of analytical methods in contrast to SDS detergent that is more likely to interfere with some commonly used analytical methods. Urea is commonly used at 8 M to ensure maximum denaturing conditions and can be dissolved in water. Cells are resuspended in the Urea solution followed by mixing during 1 hour. The extract is then clarified by centrifugation to remove the insoluble pellet.

[0130] SDS denatures proteins when heated and imparts a strong negative charge to all proteins. SDS binds strongly to proteins in the ratio of one SDS molecule per two amino acids. This makes SDS extraction a very efficient method to assess the amount of total protein, both soluble and insoluble. In general, a 2% (w/v) SDS concentration in a buffer solution between pH 6.7-7.5 is added to an equal volume of cell suspension from a cell harvest followed by mixing and heating at 95° C. for 5 minutes. Then the samples are cooled down to room temperature before centrifugation and analysis.

[0131] FIG. 3. shows the concentration of different N-intein constructs in the supernatants after extraction of proteins in the cell harvest by the use of different methods. The amount of cells and the extraction volumes were normalized prior to extraction so that the actual concentration can be directly compared. Each bar shows the average concentration for a certain construct derived from the extraction of cells from three different cell cultures. The error bars show the standard deviation. As can be seen in FIG. 3. the concentration of the protein constructs are highest in the supernatants after extraction using SDS and Urea. The relative difference between the concentration of the different constructs reflect a varying degree of expression from the different cell cultures.

Constructs A52 and B97 had the highest N-intein expression in total according to concentration in the SDS extracts, 871 and 803 µg/ml respectively. N-intein concentrations from the Urea extracts are generally lower compared with SDS extracts but follows roughly the same pattern. The interesting findings can be seen in the N-intein concentrations from the sonication and NP40 extracts where only the soluble proteins are found. A52, a construct that does not comprise the substitution mutations K24E or R25N has the lowest concentration of N-intein compared with the other constructs with 27.5 µg/ml in NP40 extracts and 37.9 µg/ml in sonicated samples. The

construct B97 comprising the K24E and R25N substitutions has a relatively high concentration of soluble N-intein in NP40 extracts, 180.3 µg/ml and in sonicated extracts, 662.3 µg/ml. This difference is more pronounced in FIG. 4., where the N-intein concentration for each respective construct and extraction method is compared with the N-intein concentration after SDS extraction of each respective construct. SDS bars are omitted since they all give the ratio 1, equal to 100%. The construct A52 lacking the mutations at position 24 and 25 has only 3 and 4% N-intein in extracts from NP40 and sonication respectively compared with SDS extracts. A single substitution, R25N in construct B83 results in a higher ratio relative to SDS extracts, 11% and 25% respectively for NP40 and sonication extracts. A single substitution, K24E in construct B82 results in a higher ratio relative to SDS extracts, 19% and 49% respectively for NP40 and sonication extracts. Construct B97, with two amino acid substitutions at position 24 and 25, K24E and R25N, results in a higher ratio relative to SDS extracts, 22% and 82% respectively for NP40 and sonication extracts. [0132] In summary, the solubility ranges achieved according to the invention in the above experiments are:

[0133] At least 10-40% soluble N-intein with a single-point mutation of R at position 25, preferred N or non-positive amino acid. At least 46-52% soluble N-intein with a single-point mutation of K at position 24, preferred E or non-positive amino acid. At least 76-88% soluble N-intein with mutations at positions 24 and 25, preferred K24E and R25N or non-positive amino acids. These values are based on Biacore CFC data on sonicated samples.

Claims

1. An N-intein protein variant derived from wildtype *Nostoc punctiforme* (Npu) or sequences having at least 95% homology therewith comprising at least one amino acid substitution of a native split intein, wherein the N-intein protein variant sequence includes a mutation in at least position 24 and/or position 25 as measured from the initial catalytic cysteine and wherein the substituted amino acid provides increased solubility in aqueous buffers compared to the native N-intein protein sequence or a consensus N-intein sequence.
2. The N-intein protein variant of claim 1 wherein the substituted amino acid(s) that provide increased solubility is a non-positive amino acid.
3. The N-intein protein variant of claim 1, wherein the substituted amino acid that provide increased solubility is K24E.
4. The N-intein protein variant of claim 1, wherein the substituted amino acid that provide increased solubility is R25N.
5. An N-intein protein variant of the wildtype N-intein domain of *Nostoc punctiforme* (Npu) wherein the wildtype Npu N-intein domain comprises the following sequence:
CLSYETEILTVEYGLLPIGKIVEKRIECTVYSVDNNGNIYTQPVAQWHDRGEQEVFE
YCLEDGSLIRATKDHKFMTVDGQMLPIDEIFERELDLMRV (SEQ ID NO 1), wherein the protein variant comprises an amino acid substitution from K to E in position 24 of SEQ ID NO 1 and R to N in position 25 of SEQ ID NO 1 to increase solubility in aqueous buffers, and wherein optionally one or more C is/are mutated to non-Cystein residues, preferably S or A.
6. The N-intein protein variant of claim 1, wherein the solubility in aqueous buffer is at least 10-40% soluble N-intein with a single-point mutation of R at position 25, preferred N or non-positive amino acid; at least 46-52% soluble N-intein with a single-point mutation of K at position 24, preferred E or non-positive amino acid; and at least 76-88% soluble N-intein with mutations at positions 24 and 25, preferred K24E and R25N or non-positive amino acids.
7. The N-intein protein variant according to claim 1, which is attached to a solid phase, such as a membrane, fiber, particle, bead or chip.
8. The N-intein protein variant sequence according to claim 7, wherein the solid phased is a chromatography resin of natural or synthetic origin.

9. The N-intein protein variant according to claim 7, wherein the solid phase is a chromatography resin, such as a natural or synthetic resin, preferably a polysaccharide such as agarose.

10. The N-intein protein variant according to claim 9, wherein the solid phase is provided with embedded magnetic particles.

11. The N-intein protein variant according to claim 9, wherein the solid phase is a non-diffusion limited resin/fibrous material.

12. The N-intein protein variant according to claim 1, wherein the N-intein is coupled to the solid phase via a Lys-tail, comprising one or more Lys, on the C-terminal.

13. The N-intein protein variant according to claim 1, wherein the N-intein is coupled to the solid phase via a Cys-tail on the C-terminal.

14. The N-intein protein variant according to claim 1, wherein 0.2-2 μ mole/ml N-intein is coupled per ml solid phase, preferably chromatography resin (ml swollen gel).

15. A split intein system comprising a N-intein protein variant according to claim 1, attached to a solid phase, and a C-intein sequence which is co-expressed with a POI (protein of interest), wherein the C-intein acts as a tag on the POI and the expressed C-intein binds to said N-intein protein variant.

16. Split intein system according to claim 15, wherein the C-intein sequence is a native split intein C-intein sequence or engineered variants thereof.

17. Split intein system according to claim 15, wherein the POI's are: proteins requiring native or near native N-terminal sequences, for example therapeutic protein candidates, biologics, antibody fragments, antibody mimetics, enzymes, recombinant proteins or peptides, such as growth factors, cytokines, chemokines, hormones, antigen (viral, bacterial, yeast, mammalian) production, vaccine production, cell surface receptors, fusion proteins.
